



# Chapter 1

# INTERGRATED-CIRCUIT

# LOGIC FAMILIES



Faculty of Computer Science and Engineering  
Department of Computer Engineering

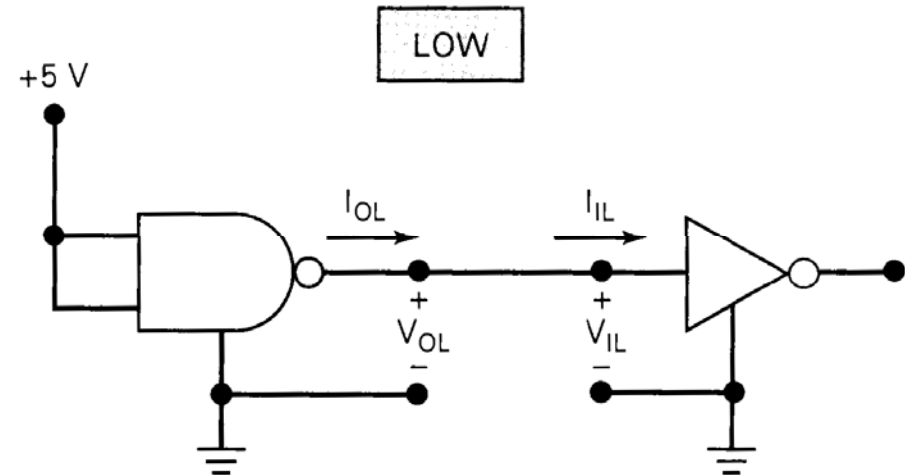
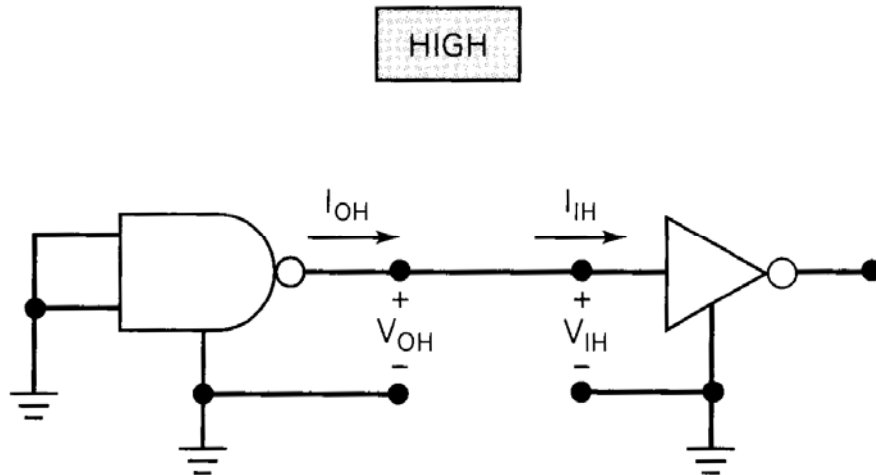
Nguyen Quang Huy

<http://www.cse.hcmut.edu.vn/~huynguyen>

# Outline

- Digital IC Terminology
- TTL Logic Family
- MOS Technology
- Open-Collector/Open-Drain Outputs
- Tristate (Three-State) Logic Outputs
- IC Interfacing

# Digital IC Terminology

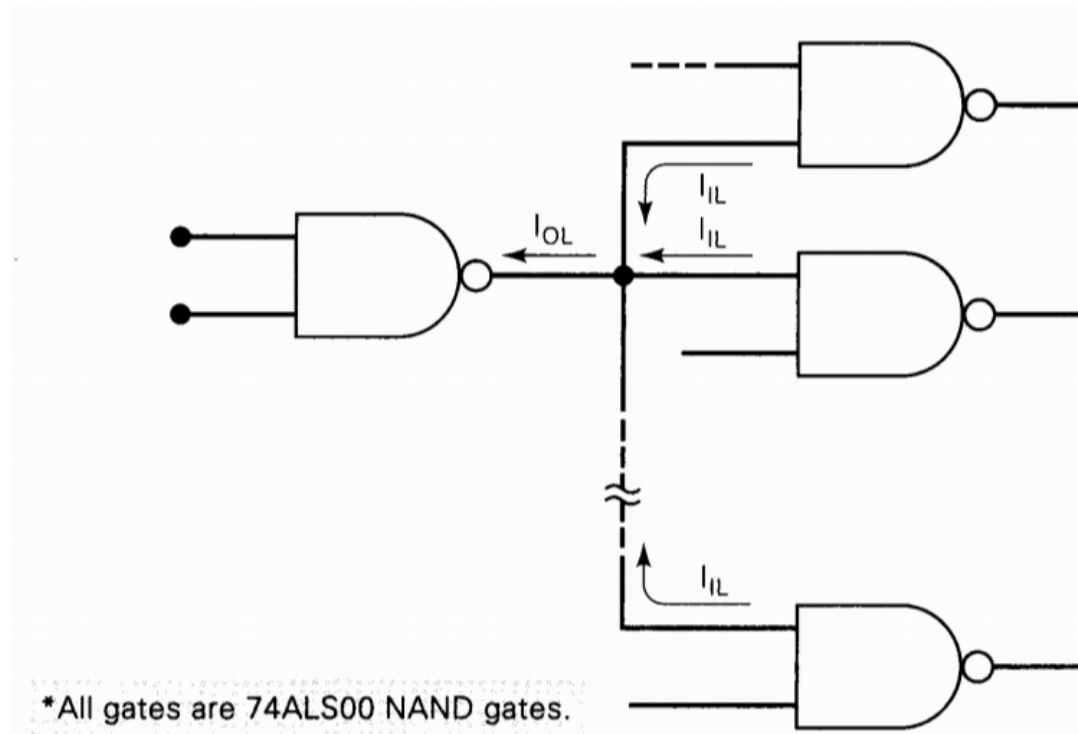


# Current and Voltage Parameters

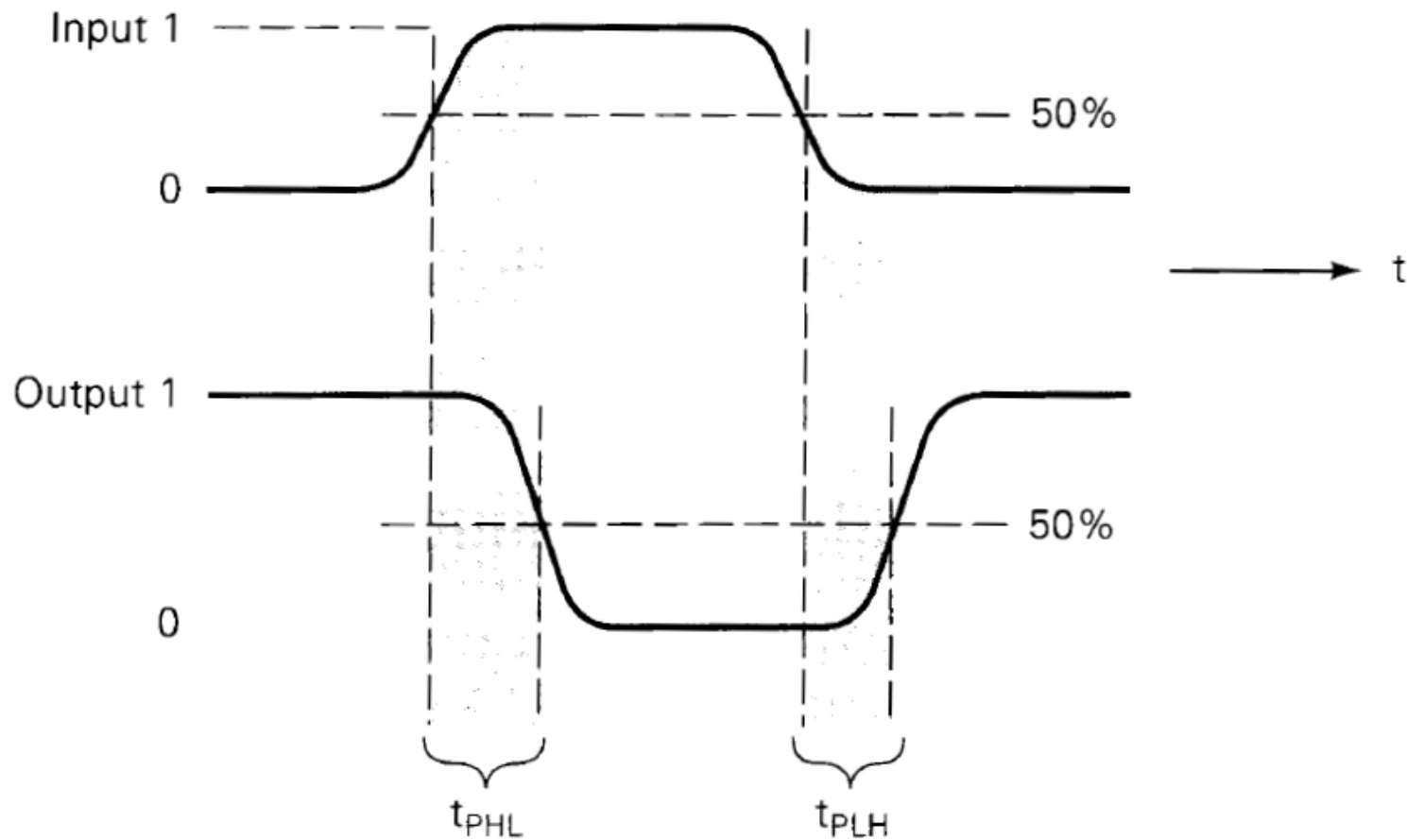
- ❑ **HIGH** logic: **1** ; **LOW** logic : **0**
- ❑  $V_{IH}(\text{min})$  – High Level Input Voltage
- ❑  $V_{IL}(\text{max})$  – Low Level Input Voltage
- ❑  $V_{OH}(\text{min})$  – High Level Output Voltage
- ❑  $V_{OL}(\text{max})$  – Low Level Output Voltage
- ❑  $I_{IH}$  – High Level Input Current
- ❑  $I_{IL}$  – Low Level Input Current
- ❑  $I_{OH}$  – High Level Output Current
- ❑  $I_{OL}$  – Low Level Output Current

# Fan-out

- Loading factor
- The maximum number of logic inputs that an output can drive reliably.

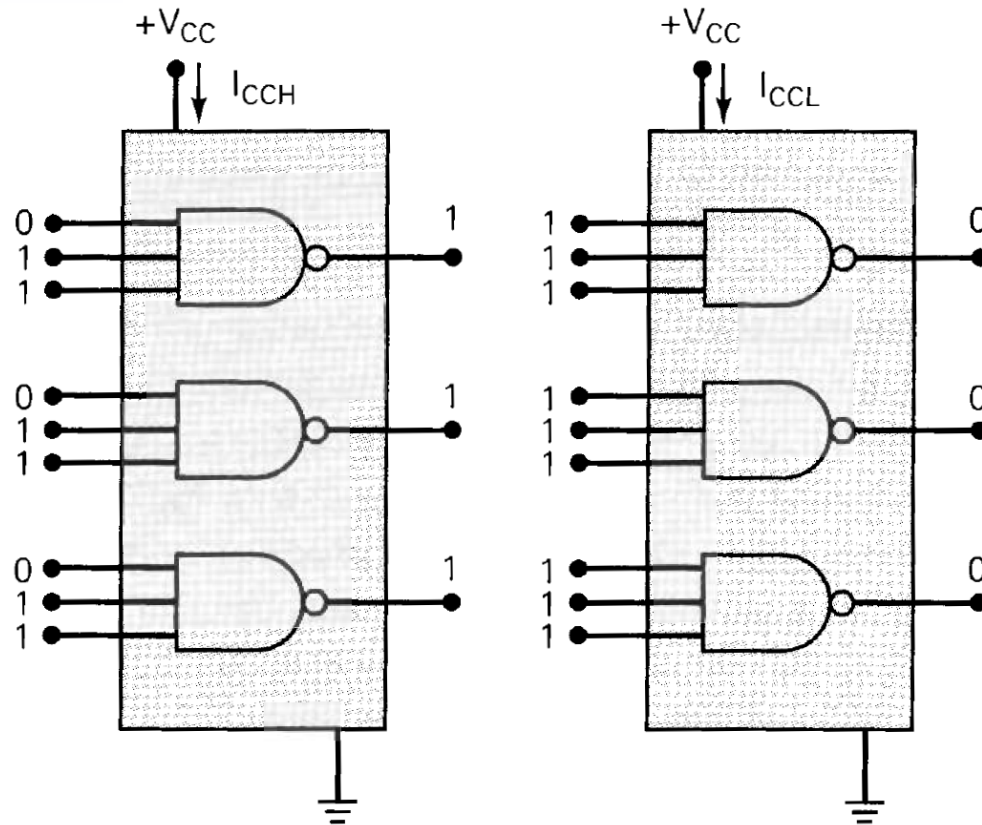


# Propagation Delays



- ❑  $t_{PLH}$ : Delay time in going from 0  $\rightarrow$  1 (LOW  $\rightarrow$  HIGH)
- ❑  $t_{PHL}$ : Delay time in going from 1  $\rightarrow$  0 (HIGH  $\rightarrow$  LOW)

# Power Requirement



- TTL devices:  $V_{CC}$
- MOS devices:  $V_{DD}$
- $I_{CCH}$ : all outputs are HIGH
- $I_{CCL}$ : all outputs are LOW

$$I_{CC(\text{avg})} = (I_{CCH} + I_{CCL}) / 2$$

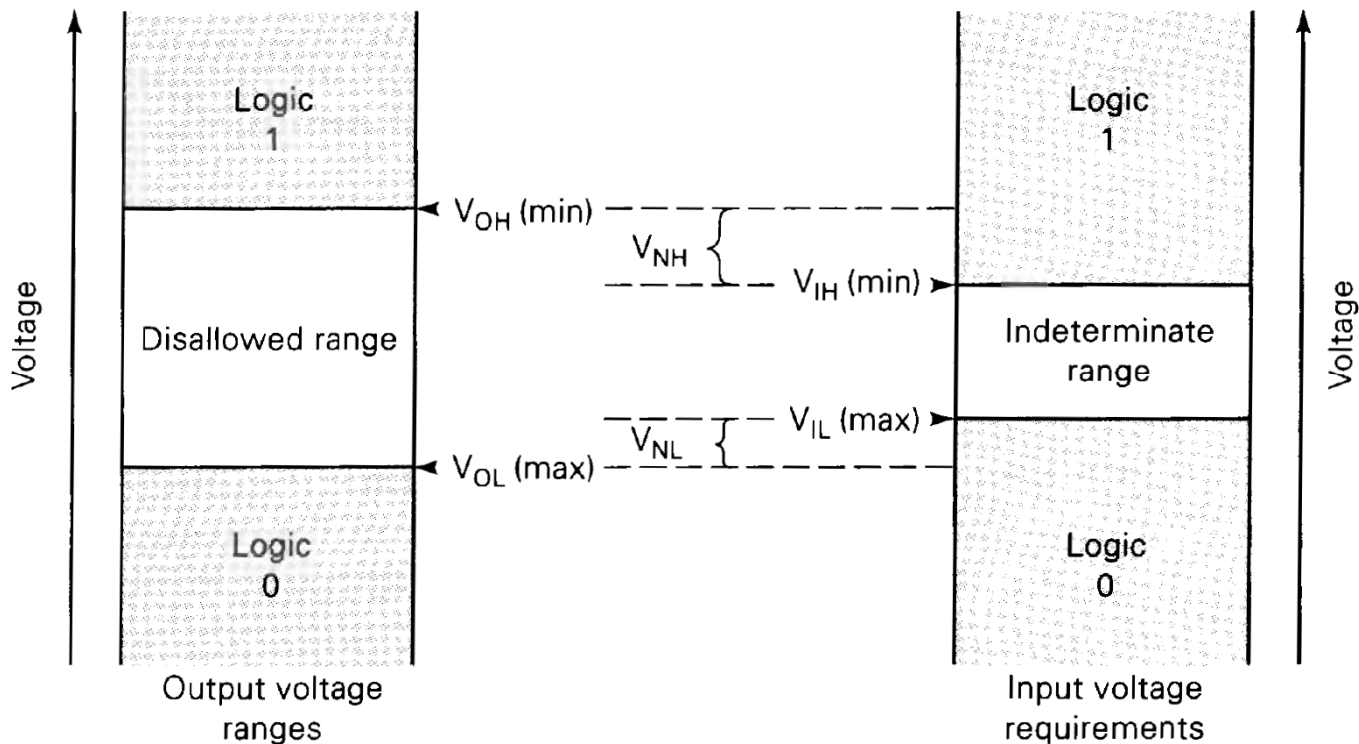
$$P_{D(\text{avg})} = I_{CC(\text{avg})} \cdot V_{CC}$$

# Speed-Power Product

- Digital IC families desire:
  - Higher speed (shorter gate propagation delay) :  $S$
  - Lower power dissipation:  $P$
  - Measuring and comparing the overall performance of an IC.

$$\text{Speed-power product} = S * P$$

# Noise Immunity



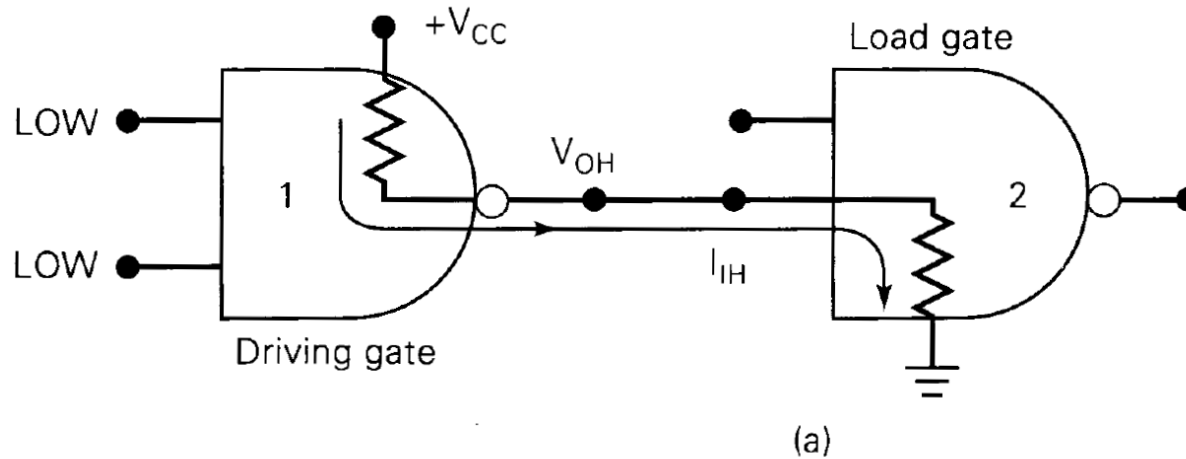
– *high-state noise margin*

$$V_{NH} = V_{OH}(\min) - V_{IH}(\min)$$

– *low-state noise margin*

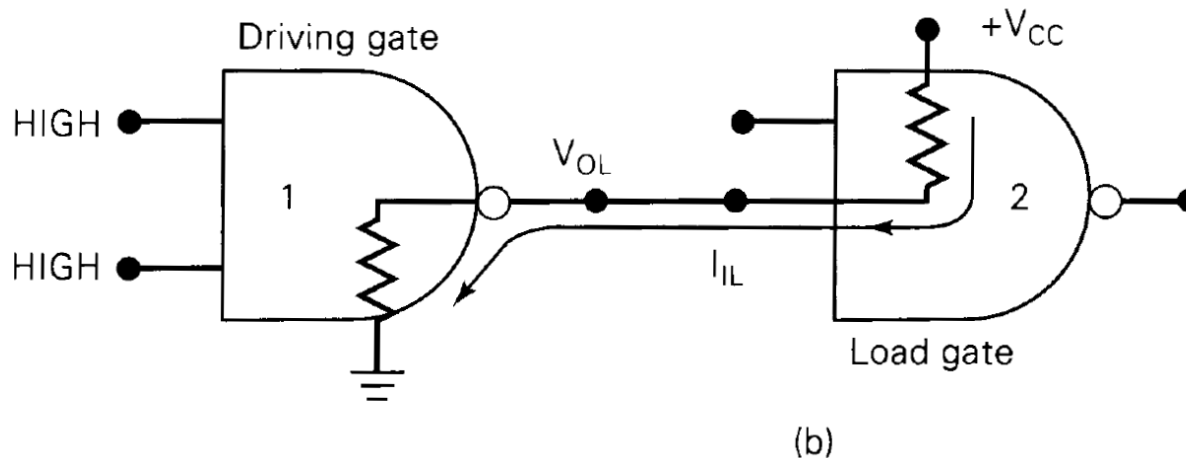
$$V_{NL} = V_{IL}(\max) - V_{OL}(\max)$$

# Current-Sourcing – Current-Sinking Logic



## Current sourcing

Driving gate supplies (sources) current to load gate in HIGH state.

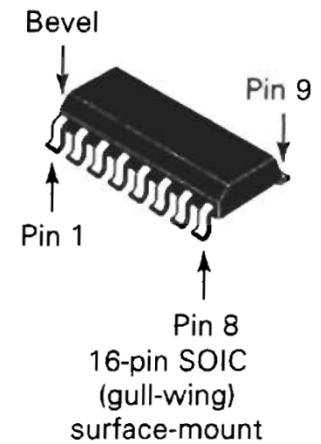
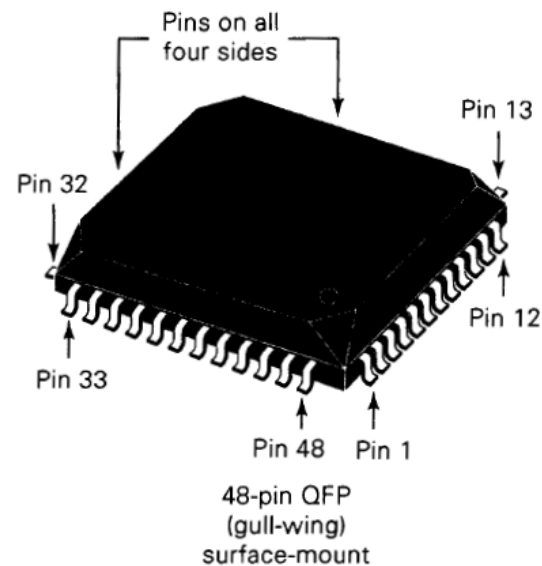
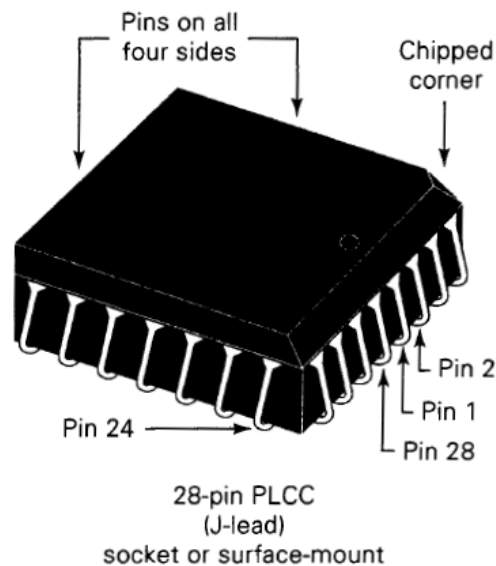
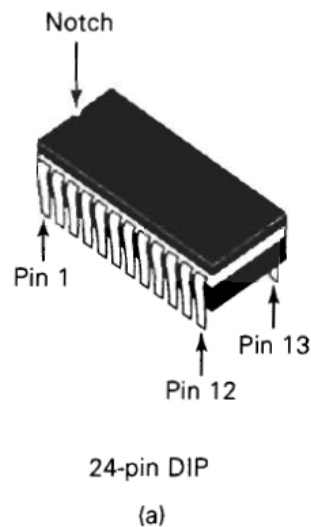


## Current sinking

Driving gate receives (sinks) current from load gate in LOW state.

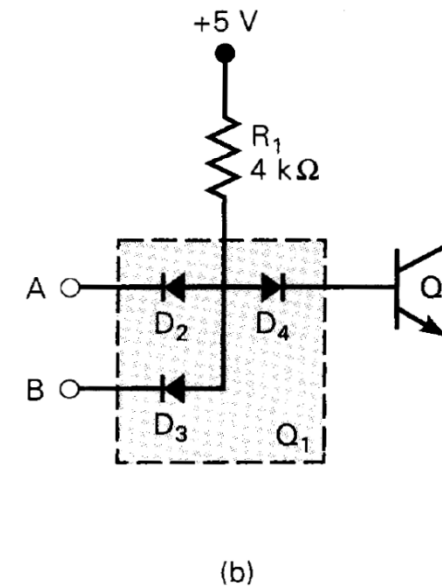
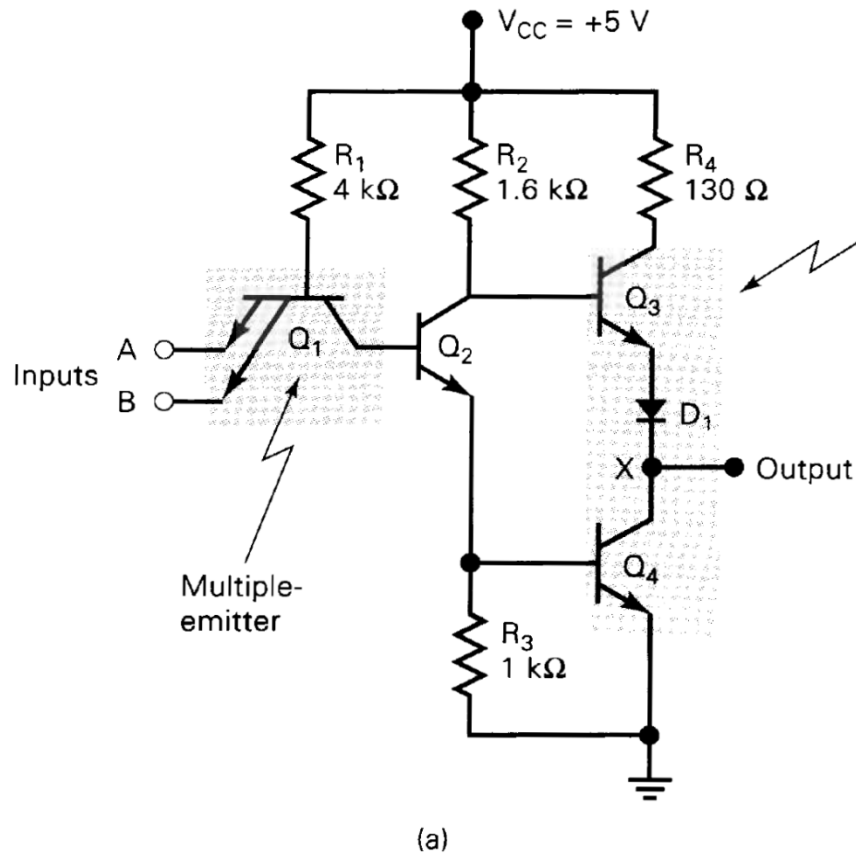
# IC Packages

- DIP (dual-in-line package)
  - Lead pitch: spacing between pins (100 mils)
- Surface-mount



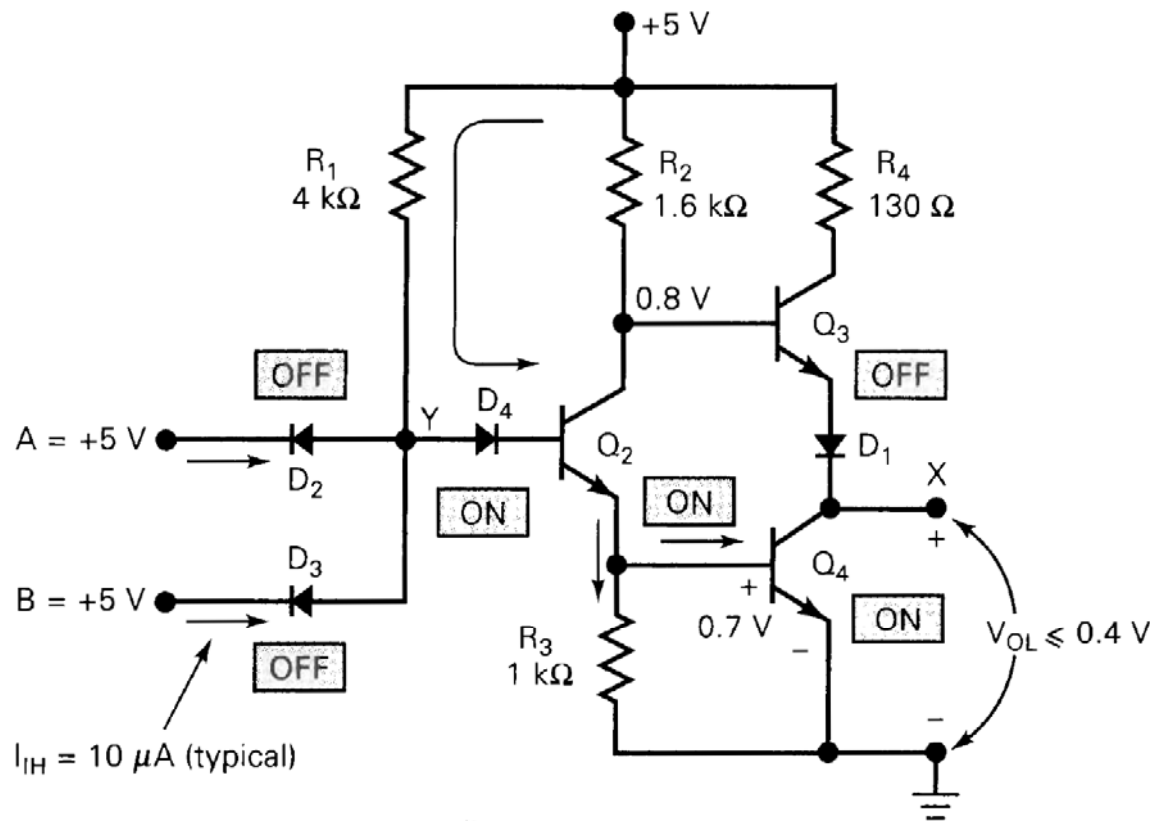
# TTL Logic Family

- TTL inputs: *multiple-emitter*
- TTL outputs: *totem-pole*



# TTL NAND Gate

- LOW State

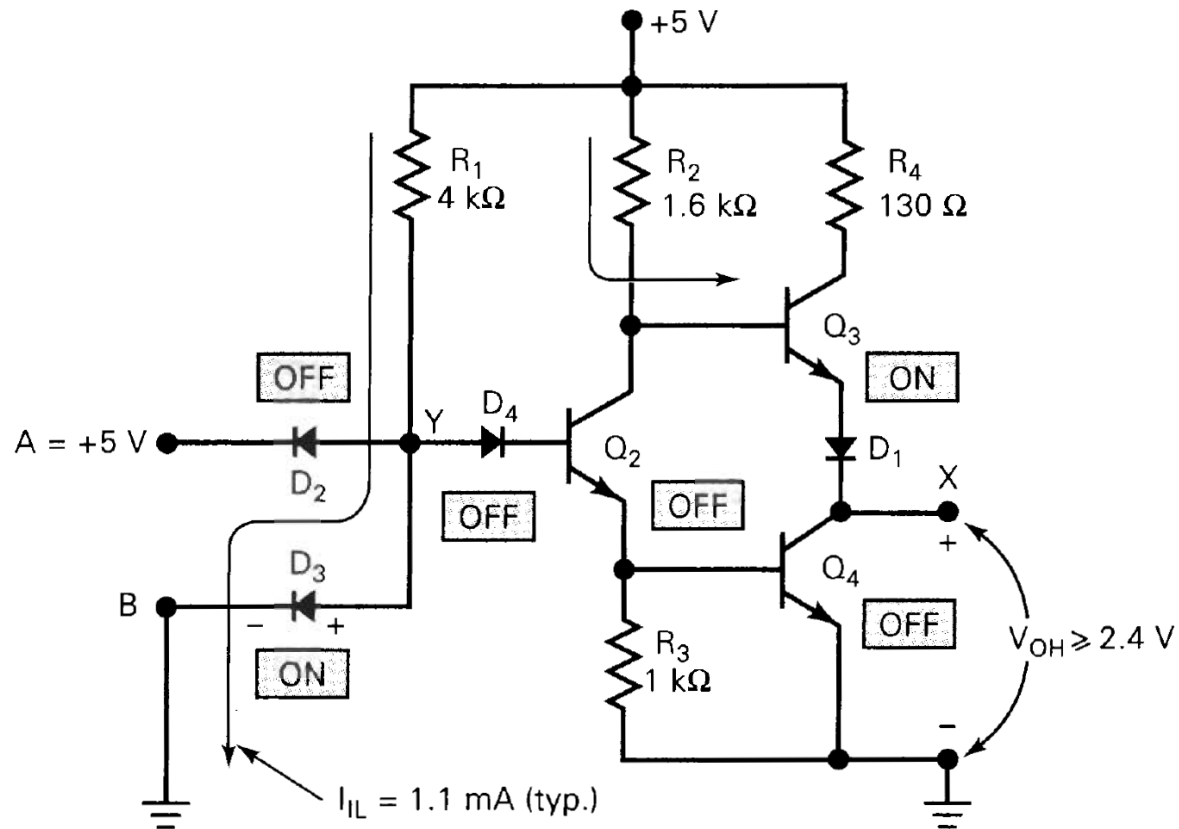


(a) LOW output

| Input conditions                                     | Output conditions                             |
|------------------------------------------------------|-----------------------------------------------|
| A and B are both HIGH ( $\geq 2$ V)                  | $Q_3$ OFF                                     |
| Input currents are very low<br>$I_{iH} = 10$ $\mu$ A | $Q_4$ ON so that $V_x$ is LOW ( $\leq 0.4$ V) |

# TTL NAND Gate

- HIGH State

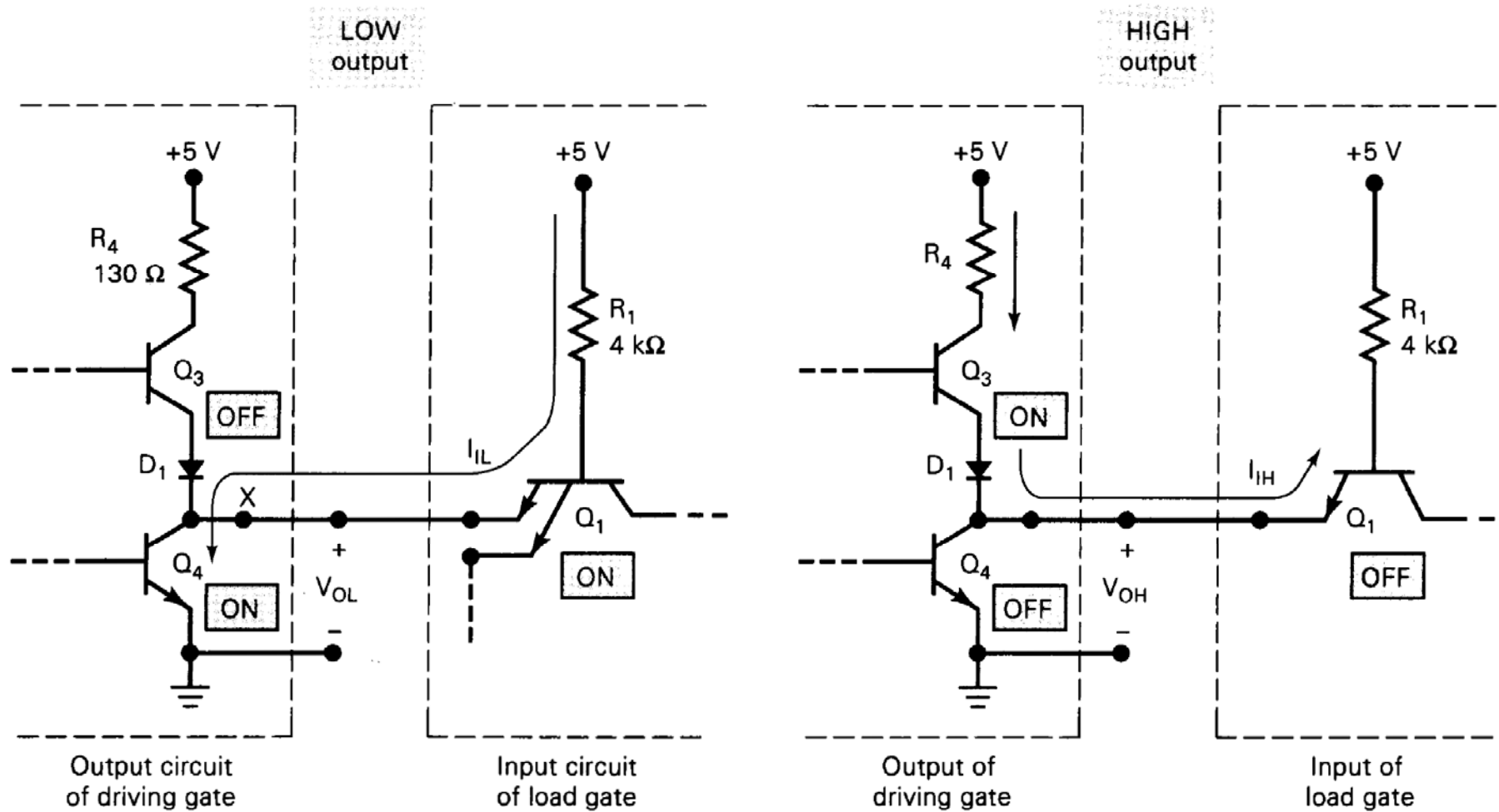


(b) HIGH output

| Input conditions                                                                     | Output conditions                                                                |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| A or B or both are LOW ( $\leq 0.8 \text{ V}$ )                                      | $Q_4$ OFF                                                                        |
| Current flows back to ground through LOW input terminal<br>$I_{IL} = 1.1 \text{ mA}$ | $Q_3$ acts as emitter-follower and $V_{OH} \geq 2.4 \text{ V}$ , typically 3.6 V |

# TTL NAND Gate

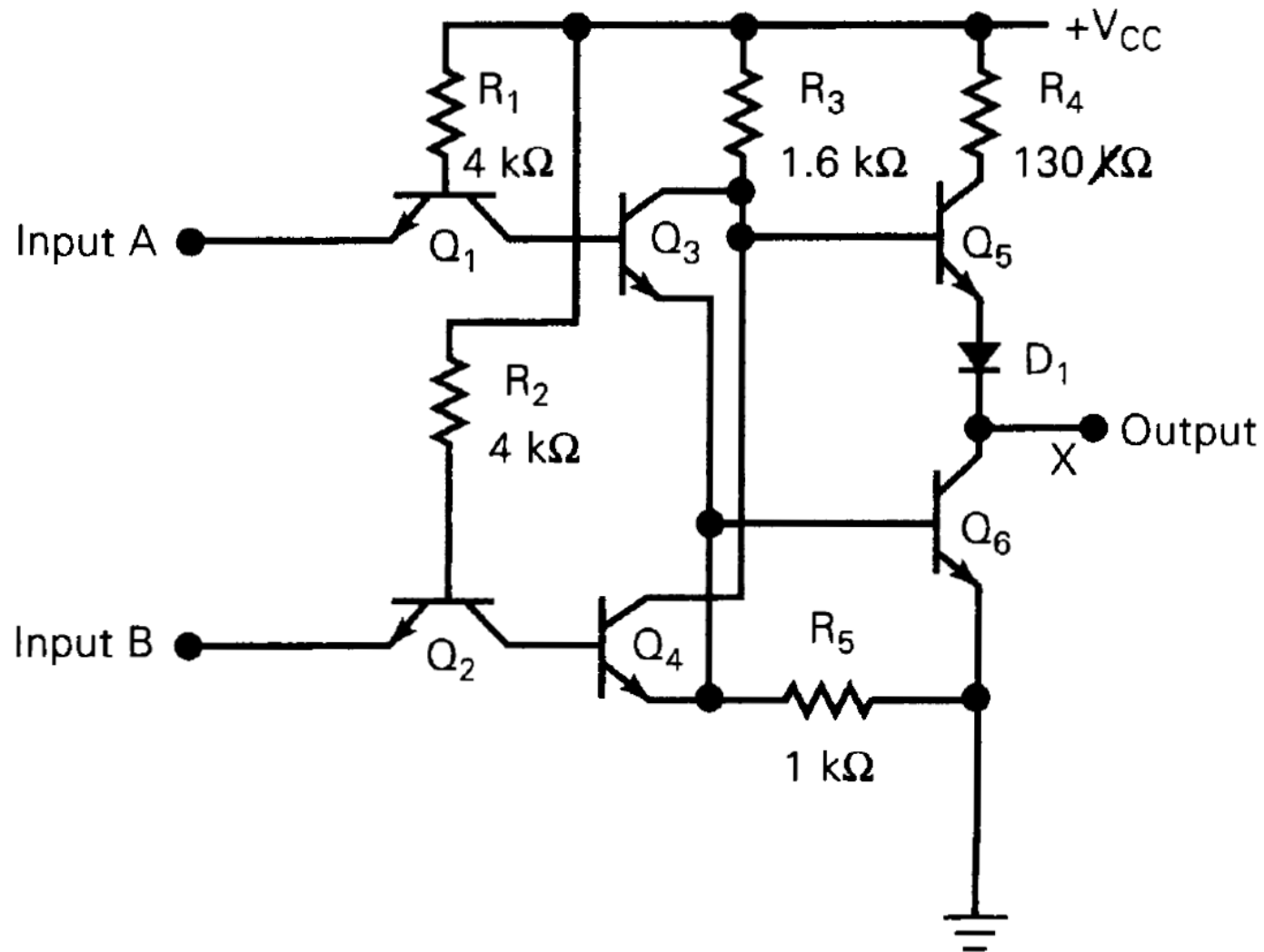
- Current-Sinking and Current Sourcing



# TTL NAND Gate

- Totem-Pole Output
  - Advantage
    - Low power dissipation
    - Fast rise-time
  - Disadvantage
    - Large current spike during switching from LOW to HIGH

# TTL NOR Gate



# TTL Series

- Standard TTL, 74
- Schottky TTL, 74S
- Low-Power Schottky TTL, 74LS (LS-TTL)
- Advanced Schottky TTL, 74AS (AS-TTL)
- Advanced Low-Power Schottky TTL, 74ALS
- 74F-Fast TTL

# TTL Data Sheets

- Supply Voltage – Temperature Range
- Voltage Levels
  - $V_{IL}, V_{OL}, V_{OH}, V_{IH} \rightarrow V_{NL}, V_{NH}$
- Maximum Voltage Ratings
- Power Dissipation
  - $I_{CCH}, I_{CCL} \rightarrow I_{CC}(\text{avg}) \rightarrow P_D(\text{avg})$
- Propagation Delays
  - $t_{PLH}, t_{PHL} \rightarrow t_{pd}(\text{avg})$

## TTL Data Sheets

|                  |                                                             | SN54ALS00A             |                                                                                                                         |      | SN74ALS00A |                     |      | UNIT |      |
|------------------|-------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------|------|------------|---------------------|------|------|------|
|                  |                                                             | MIN                    | NOM                                                                                                                     | MAX  | MIN        | NOM                 | MAX  |      |      |
| V <sub>CC</sub>  | Supply voltage                                              | 4.5                    | 5                                                                                                                       | 5.5  | 4.5        | 5                   | 5.5  | V    |      |
| V <sub>IH</sub>  | High-level input voltage                                    | 2                      |                                                                                                                         |      | 2          |                     |      | V    |      |
| V <sub>IL</sub>  | Low-level input voltage                                     |                        |                                                                                                                         | 0.8‡ |            |                     | 0.8  | V    |      |
|                  |                                                             |                        |                                                                                                                         | 0.7§ |            |                     |      |      |      |
| I <sub>OH</sub>  | High-level output current                                   |                        |                                                                                                                         | −0.4 |            |                     | −0.4 | mA   |      |
| I <sub>OL</sub>  | Low-level output current                                    |                        |                                                                                                                         | 4    |            |                     | 8    | mA   |      |
| T <sub>A</sub>   | Operating free-air temperature                              | −55                    |                                                                                                                         | 125  | 0          |                     | 70   | °C   |      |
| PARAMETER        | TEST CONDITIONS                                             |                        | SN54ALS00A                                                                                                              |      |            | SN74ALS00A          |      |      | UNIT |
|                  |                                                             |                        | MIN                                                                                                                     | TYP† | MAX        | MIN                 | TYP† | MAX  |      |
| V <sub>IK</sub>  | V <sub>CC</sub> = 4.5 V, I <sub>I</sub> = −18 mA            |                        |                                                                                                                         |      | −1.2       |                     | −1.5 | V    |      |
| V <sub>OH</sub>  | V <sub>CC</sub> = 4.5 V to 5.5 V, I <sub>OH</sub> = −0.4 mA |                        | V <sub>CC</sub> − 2                                                                                                     |      |            | V <sub>CC</sub> − 2 |      |      | V    |
| V <sub>OL</sub>  | V <sub>CC</sub> = 4.5 V                                     | I <sub>OL</sub> = 4 mA |                                                                                                                         | 0.25 | 0.4        |                     | 0.25 | 0.4  | V    |
|                  |                                                             | I <sub>OL</sub> = 8 mA |                                                                                                                         |      |            |                     | 0.35 | 0.5  |      |
| I <sub>I</sub>   | V <sub>CC</sub> = 5.5 V, V <sub>I</sub> = 7 V               |                        |                                                                                                                         |      | 0.1        |                     | 0.1  | mA   |      |
| I <sub>IH</sub>  | V <sub>CC</sub> = 5.5 V, V <sub>I</sub> = 2.7 V             |                        |                                                                                                                         |      | 20         |                     | 20   | μA   |      |
| I <sub>IL</sub>  | V <sub>CC</sub> = 5.5 V, V <sub>I</sub> = 0.4 V             |                        |                                                                                                                         |      | −0.1       |                     | −0.1 | mA   |      |
| I <sub>O</sub> ‡ | V <sub>CC</sub> = 5.5 V, V <sub>O</sub> = 2.25 V            |                        | −20                                                                                                                     |      | −112       | −30                 |      | −112 | mA   |
| I <sub>CCH</sub> | V <sub>CC</sub> = 5.5 V, V <sub>I</sub> = 0                 |                        |                                                                                                                         | 0.5  | 0.85       |                     | 0.5  | 0.85 | mA   |
| I <sub>CCL</sub> | V <sub>CC</sub> = 5.5 V, V <sub>I</sub> = 4.5 V             |                        |                                                                                                                         | 1.5  | 3          |                     | 1.5  | 3    | mA   |
| PARAMETER        | FROM<br>(INPUT)                                             | TO<br>(OUTPUT)         | V <sub>CC</sub> = 4.5 V to 5.5 V,<br>C <sub>L</sub> = 50 pF,<br>R <sub>L</sub> = 500 Ω,<br>T <sub>A</sub> = MIN to MAX§ |      |            |                     | UNIT |      |      |
|                  |                                                             |                        | SN54ALS00A                                                                                                              |      | SN74ALS00A |                     |      |      |      |
|                  |                                                             |                        | MIN                                                                                                                     | MAX  | MIN        | MAX                 |      |      |      |
| t <sub>PLH</sub> | A or B                                                      | Y                      | 3                                                                                                                       | 15   | 3          | 11                  | ns   |      |      |
| t <sub>PHL</sub> |                                                             |                        | 2                                                                                                                       | 9    | 2          | 8                   |      |      |      |

# TTL Data Sheets

- Loading and Fan-out

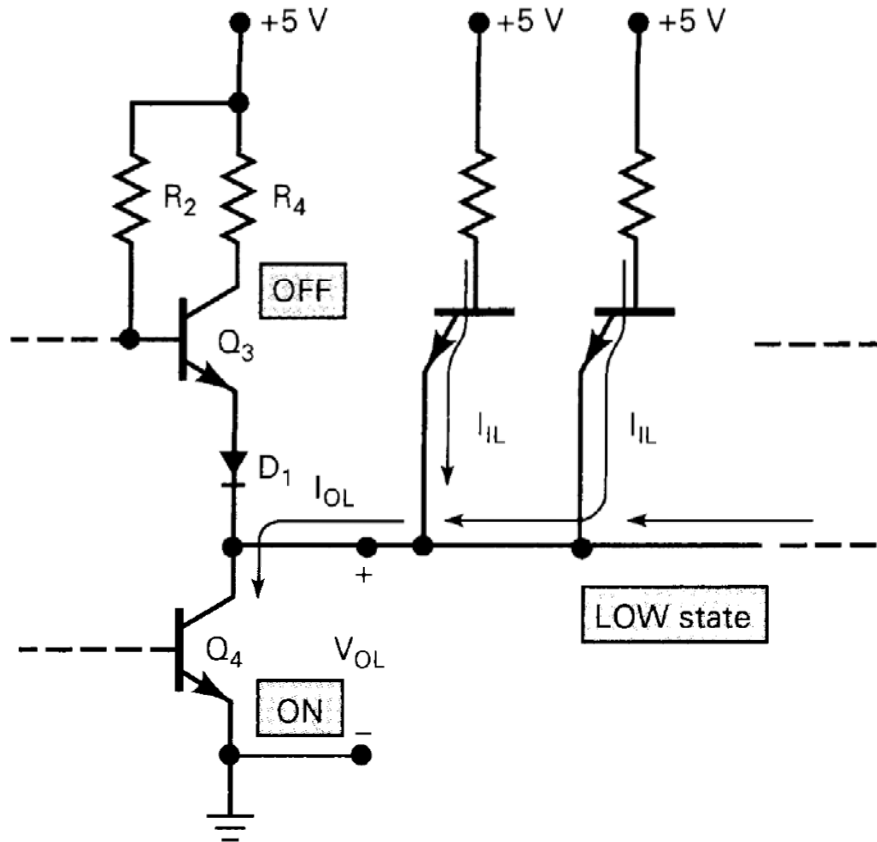
$$\text{fan-out(LOW)} = \frac{I_{OL}(\text{max})}{I_{IL}(\text{max})}$$

$$\text{fan-out(HIGH)} = \frac{I_{OH}(\text{max})}{I_{IH}(\text{max})}$$

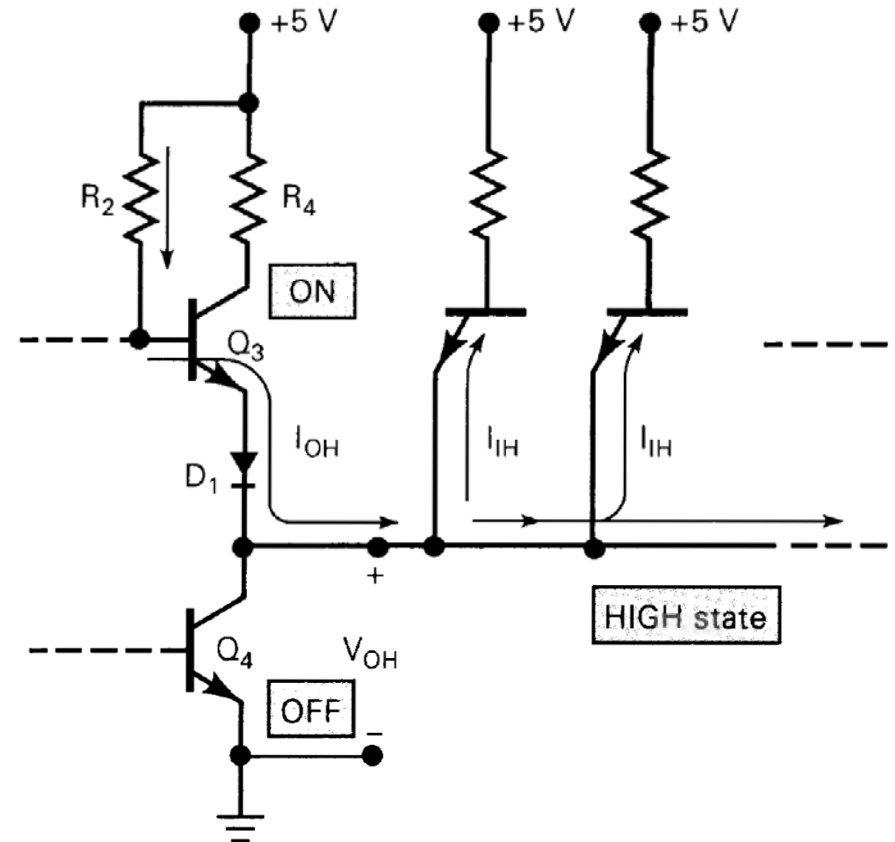
- Example: In datasheet of 74ALS00
  - $I_{OL}(\text{max}) = 8 \text{ mA}$ ,  $I_{IL}(\text{max}) = 0.1 \text{ mA}$
  - $I_{OH}(\text{max}) = 0.4 \text{ mA} = 400 \text{ }\mu\text{A}$ ,  $I_{OL} = 20 \text{ }\mu\text{A}$
  - **fan-out(LOW) = 80, fan-out(HIGH) = 20**

# TTL Data Sheets

- Loading and Fan-out



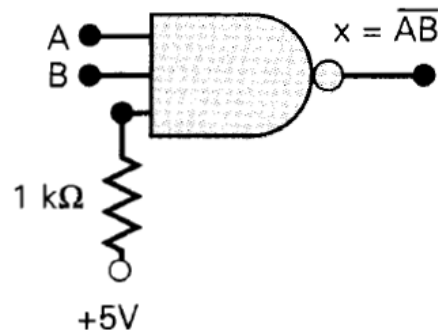
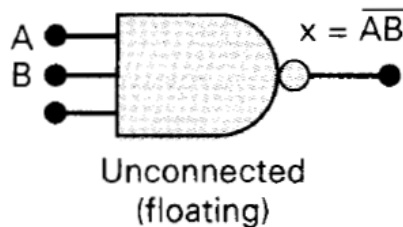
$$I_{OL} = \sum I_{IL}$$



$$I_{OH} = \sum I_{IH}$$

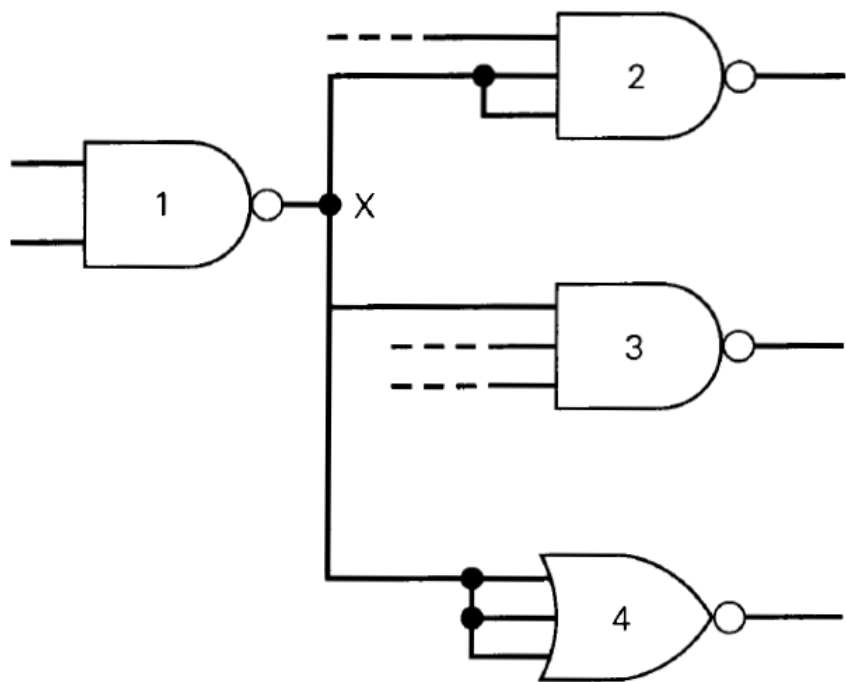
# Other TTL Characteristics

- Unconnected Inputs (Floating)
  - Open input acts exactly like a logical 1 (HIGH)
- Unused Inputs
  - Left disconnected (undesirable – antenna)
  - Connect to  $V_{CC}$  (+5V) through a 1-k $\Omega$  resistor or to GND to produce for a constant-output logic level
  - Tied to a used input



# Other TTL Characteristics

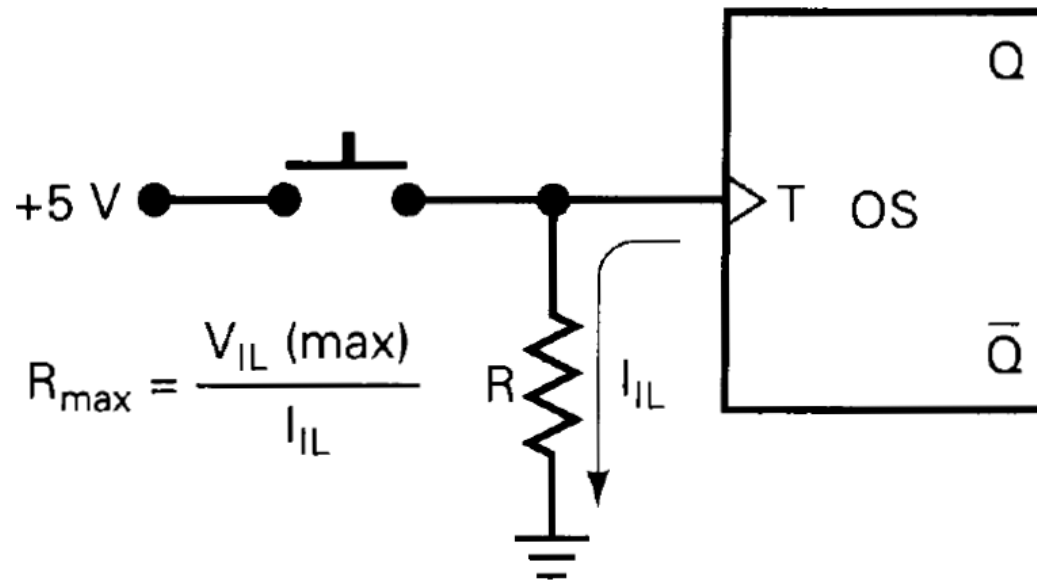
- Tied-Together Inputs
  - Represent a load – **Sum of the load current rating**
  - NAND, AND gates: LOW-state input load – **single input**



| Loading on gate 1 output |       |              |       |  |
|--------------------------|-------|--------------|-------|--|
| HIGH                     |       | LOW          |       |  |
| Load Current             | Gate  | Load Current | Gate  |  |
| 40 $\mu$ A               | 2     | 0.4 mA       | 2     |  |
| 20 $\mu$ A               | 3     | 0.4 mA       | 3     |  |
| 60 $\mu$ A               | 4     | 1.2 mA       | 4     |  |
| 120 $\mu$ A              | Total | 2.0 mA       | Total |  |

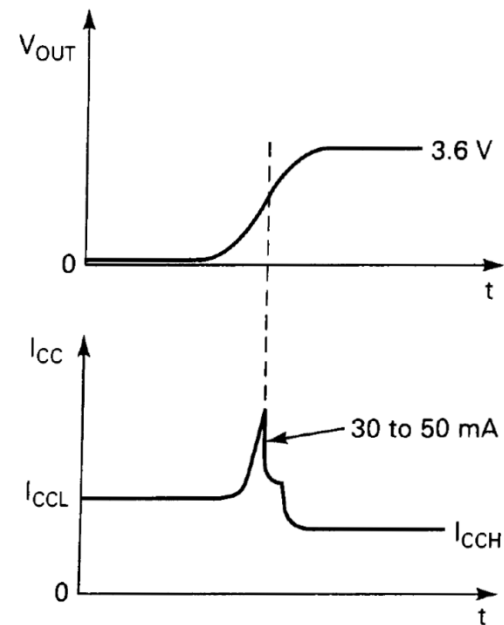
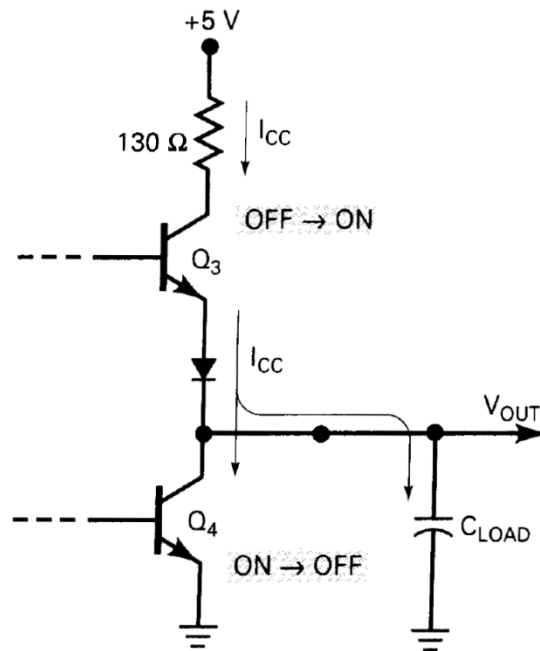
# Other TTL Characteristics

- Biasing TTL Inputs Low



# Other TTL Characteristics

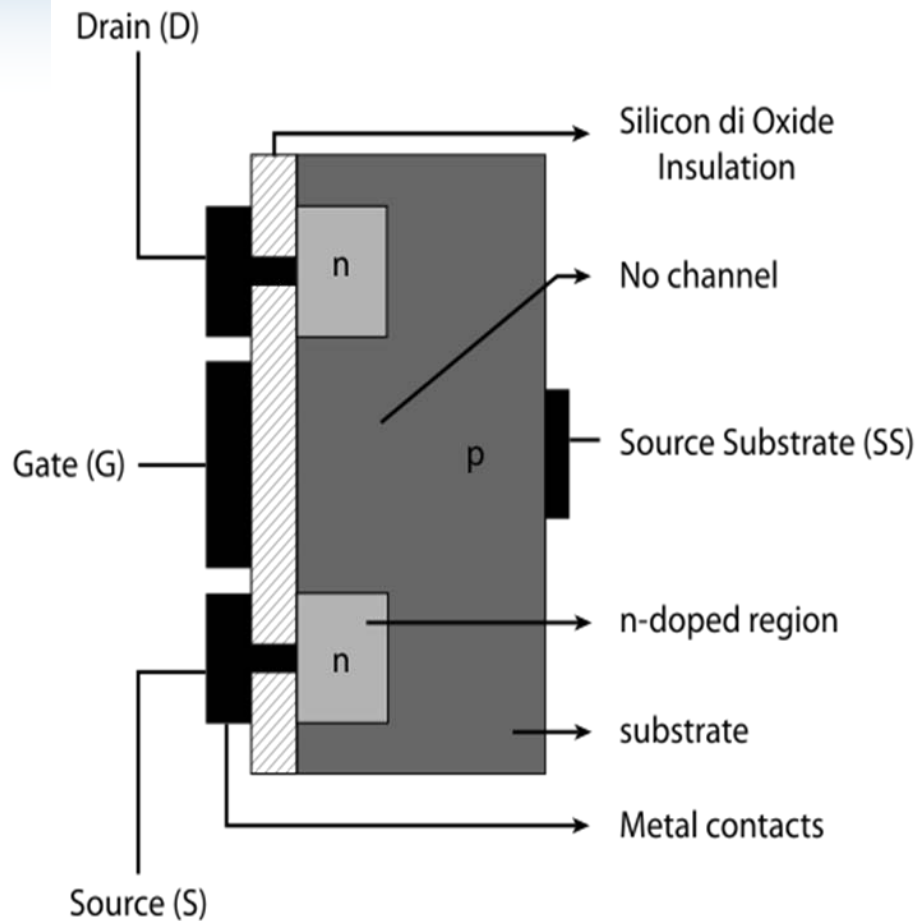
- Current Transients
  - high-amplitude current spike occurs when a totem-pole TTL output goes from LOW to HIGH
  - **Power-supply decoupling:** capacitors connect from  $V_{CC}$  to GND ( $0.01 \mu\text{F}$  or  $0.1 \mu\text{F}$ )



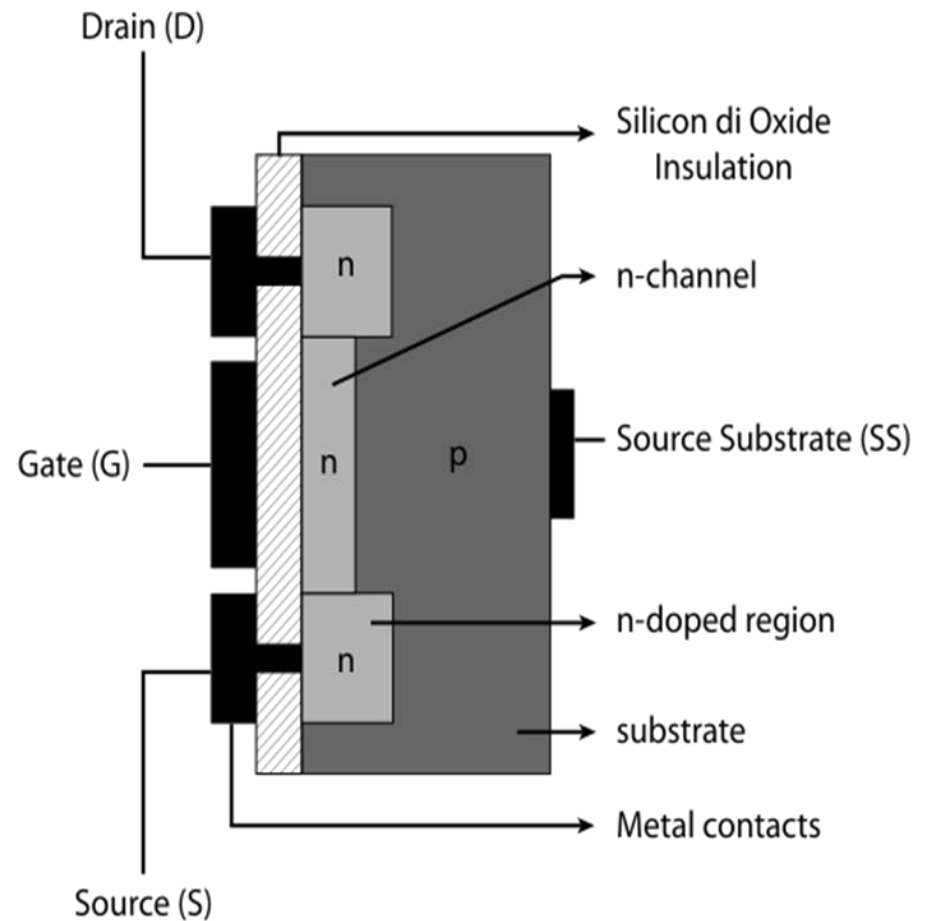
# MOSFET

- Metal-oxide-semiconductor field-effect transistor
- Advantage
  - Simple, inexpensive to fabricate
  - Small, little power consumption
  - Suited for complex ICs
  - Faster than 74, 74LS, 74ALS TTL Series
- Disadvantage
  - susceptibility to static-electricity damage
- Depletion and enhancement MOSFET
  - [http://www.electronics-tutorials.ws/transistor/tran\\_6.html](http://www.electronics-tutorials.ws/transistor/tran_6.html)

# MOSFET Structure



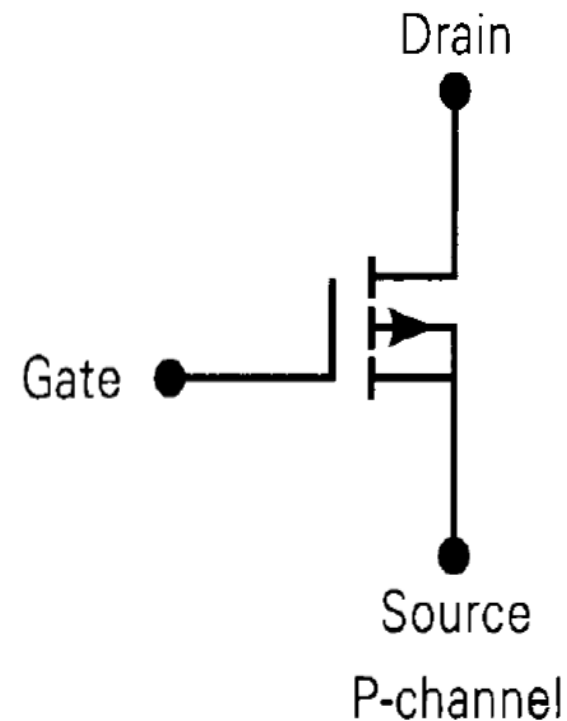
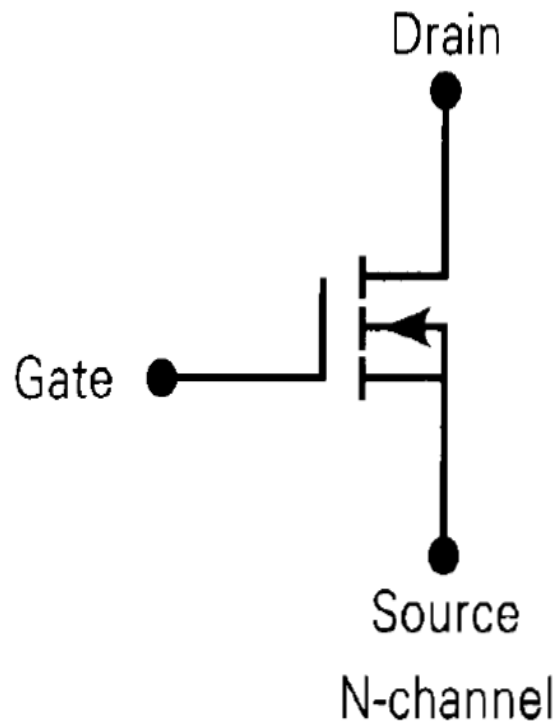
NMOS without channel  
formed (**OFF state**)



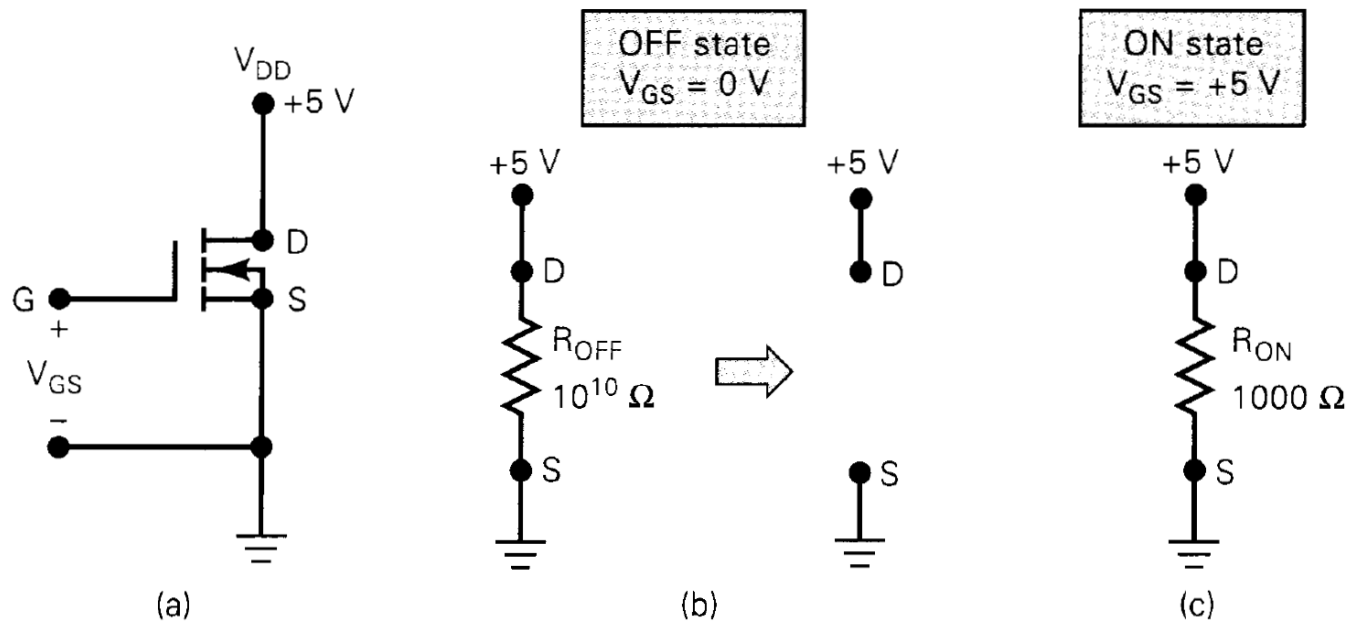
NMOS with channel  
formed (**ON state**)

# Enhancement MOSFET

- Use in digital ICs
- Enhancement MOSFET: normally “open” (OFF)



# Basic MOSFET Switch

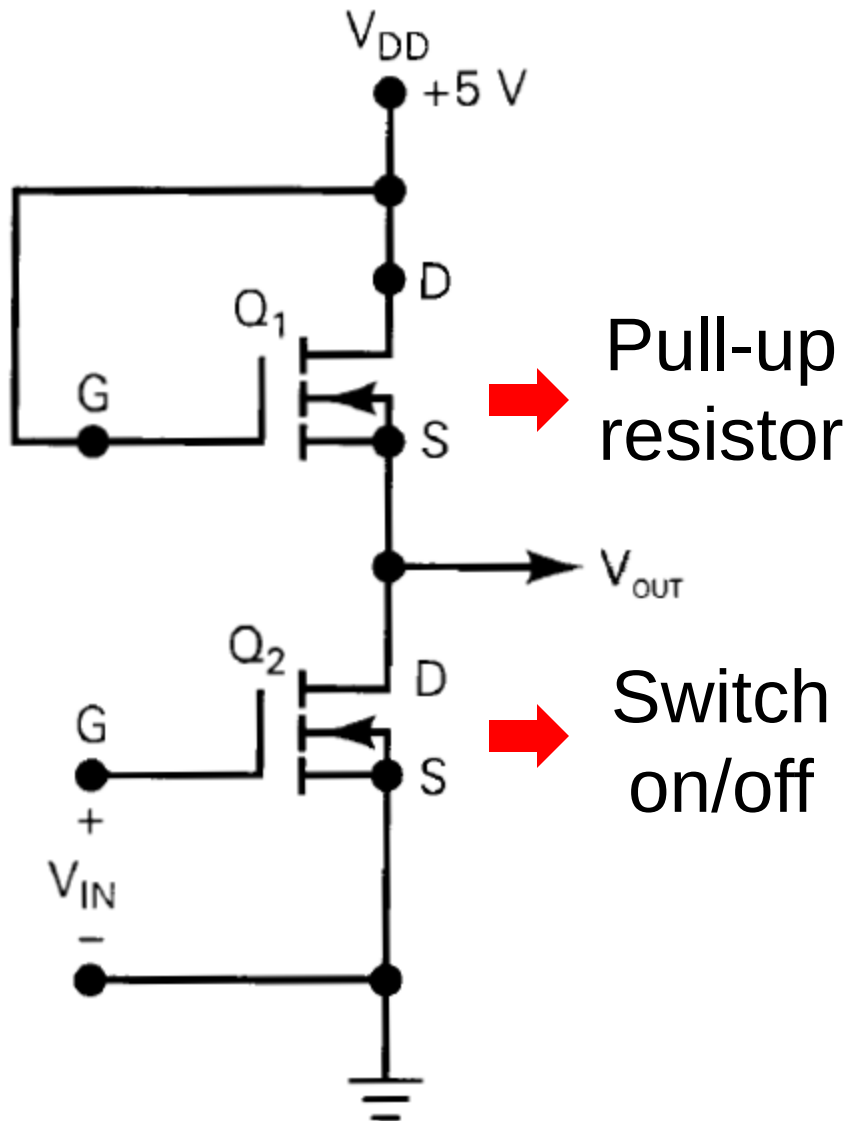


|           | Drain-to-Source Bias | Gate-to-Source Voltage ( $V_{GS}$ ) Needed for Conduction | $R_{ON} (\Omega)$ | $R_{OFF} (\Omega)$ |
|-----------|----------------------|-----------------------------------------------------------|-------------------|--------------------|
| P-channel | Negative             | Typically more negative than $-1.5\text{ V}$              | 1000 (typical)    | $10^{10}$          |
| N-channel | Positive             | Typically more positive than $+1.5\text{ V}$              | 1000 (typical)    | $10^{10}$          |

# Digital MOSFET Circuit

- N-MOS or P-MOS
  - Use MOSFET as a switch
  - Implement all resistors using channel resistance of a MOSFET
  - Simple circuits and fabrication processes
- C-MOS (complementary MOS)
  - Use both P- and N-channel MOSFET
  - Increase complexity of IC fabrication process, lower packing density
  - Faster and less power consumption

# N-MOS Inverter

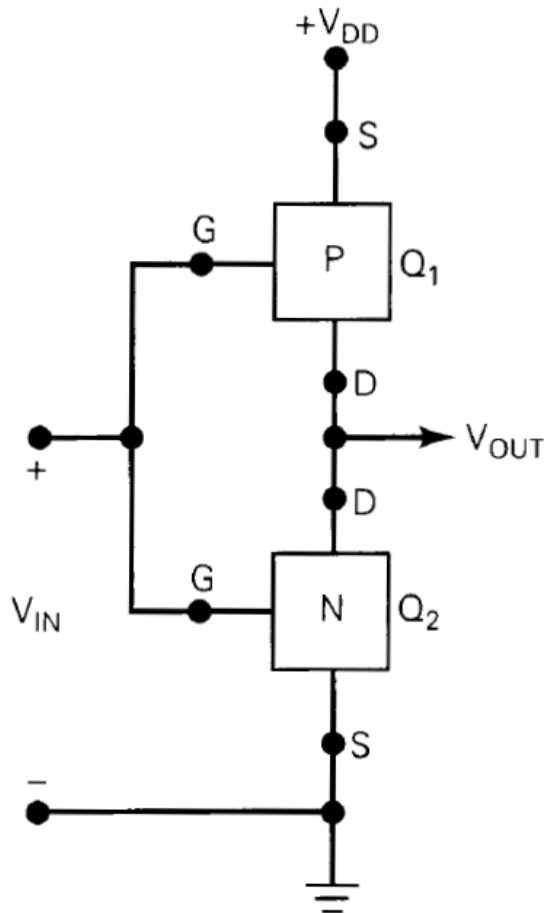


| $V_{IN}$          | $Q_1$                         | $Q_2$                             | $V_{OUT} = \overline{V_{IN}}$ |
|-------------------|-------------------------------|-----------------------------------|-------------------------------|
| 0 V<br>(logic 0)  | $R_{ON} = 100\text{ k}\Omega$ | $R_{OFF} = 10^{10}\text{ }\Omega$ | +5 V<br>(logic 1)             |
| +5 V<br>(logic 1) | $R_{ON} = 100\text{ k}\Omega$ | $R_{ON} = 1\text{ k}\Omega$       | +0.05 V<br>(logic 0)          |

- Gate of  $Q_1$  connected to the drain  $\rightarrow$  always ON  
 $\rightarrow$  pull-up resistor  $100\text{ k}\Omega$

# Complementary MOS Logic

- CMOS Inverter

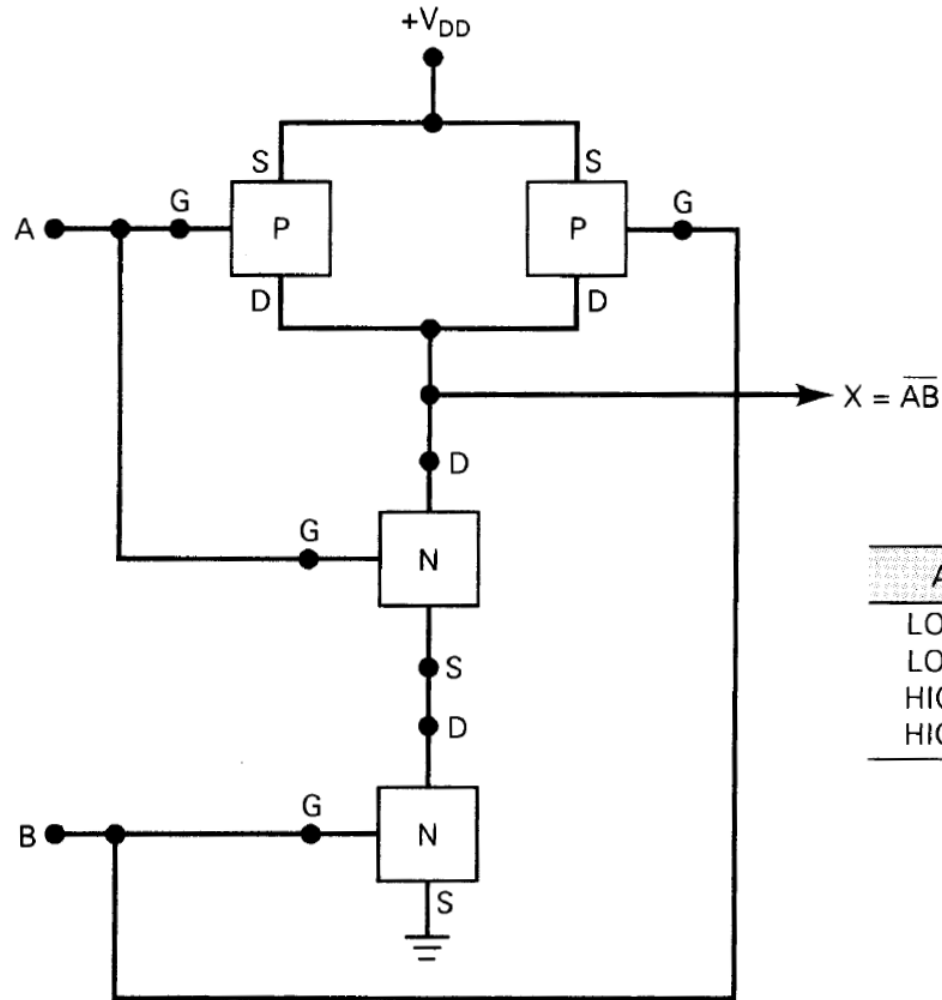


| $V_{IN}$                   | $Q_1$                              | $Q_2$                              | $V_{OUT}$             |
|----------------------------|------------------------------------|------------------------------------|-----------------------|
| $+V_{DD}$<br>(logic 1)     | OFF<br>$R_{OFF} = 10^{10} \Omega$  | ON<br>$R_{ON} = 1 \text{ k}\Omega$ | $\approx 0 \text{ V}$ |
| $0 \text{ V}$<br>(logic 0) | ON<br>$R_{ON} = 1 \text{ k}\Omega$ | OFF<br>$R_{OFF} = 10^{10} \Omega$  | $\approx +V_{DD}$     |

$$V_{OUT} = \overline{V_{IN}}$$

# Complementary MOS Logic

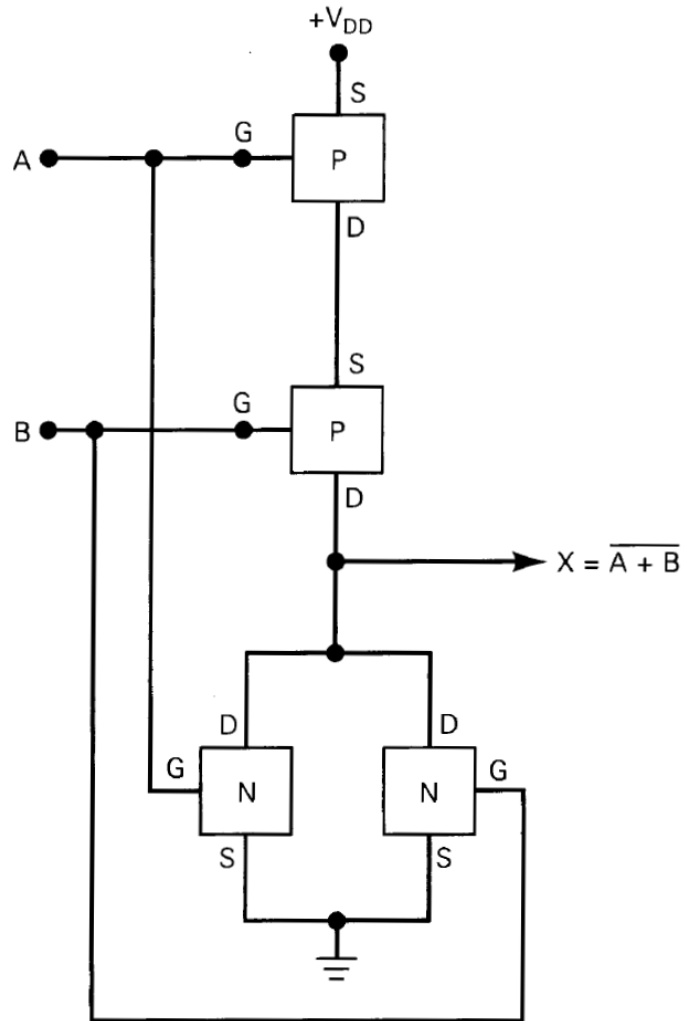
- CMOS NAND Gate



| A    | B    | X    |
|------|------|------|
| LOW  | LOW  | HIGH |
| LOW  | HIGH | HIGH |
| HIGH | LOW  | HIGH |
| HIGH | HIGH | LOW  |

# Complementary MOS Logic

- CMOS NOR Gate



| A    | B    | X    |
|------|------|------|
| LOW  | LOW  | HIGH |
| LOW  | HIGH | LOW  |
| HIGH | LOW  | LOW  |
| HIGH | HIGH | LOW  |

# CMOS Series

- 4000/14000 Series
- 74C Series
- 74HC/HCT (*High-Speed CMOS*)
- 74AC/ACT (*Advanced CMOS*)
- 74AHC/74AHCT (*Advanced High-Speed CMOS*)
- BICMOS 5-Volt Logic

# Low-Voltage Technology

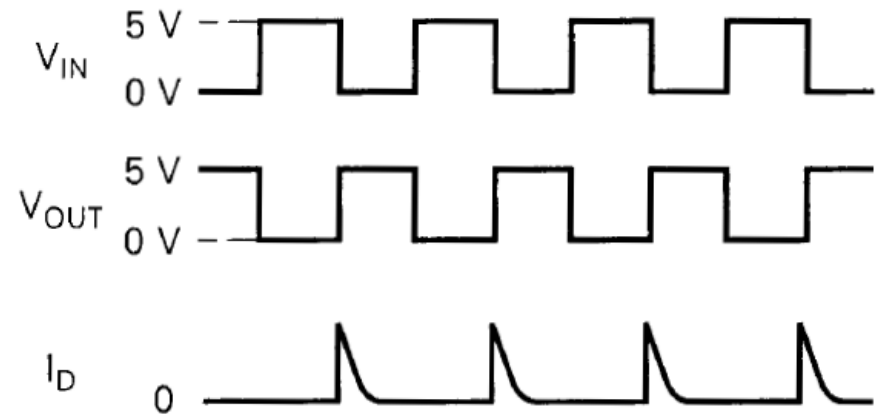
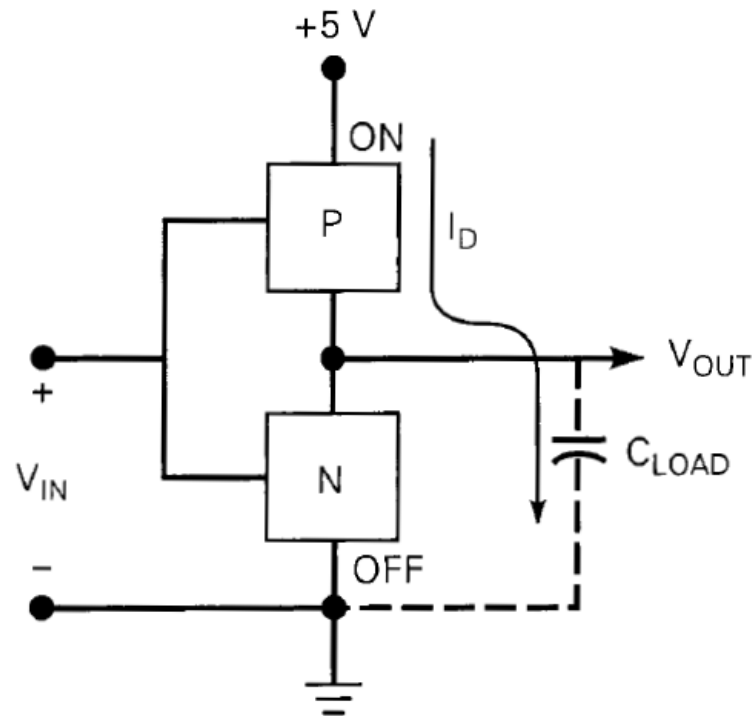
- CMOS Family
  - 74LVC (*Low-Voltage CMOS*)
  - 74ALVC (*Advanced Low-Voltage CMOS*)
  - 74LV (*Low-Voltage*)
  - 74AVC (*Advanced Very-Low-Voltage CMOS*)
- BiCMOS Family
  - 74LVT (*Low-Voltage BiCMOS Technology*)
  - 74LVT (*Advanced Low-Voltage BiCMOS Technology*)
  - 74ALB (*Advanced Low-Voltage BiCMOS*)

# CMOS Characteristics (1)

- Power-Supply Voltage
- Logic Voltage Levels
  - Not sink/source any significant amount of current
  - $V_{OL}$ : close to 0 V,  $V_{OH}$ : close to VDD
- Noise Margins: greater than TTL
  - $V_{NH} = V_{OH}(\text{min}) - V_{IH}(\text{min})$
  - $V_{NL} = V_{IL}(\text{max}) - V_{OL}(\text{max})$
- Power Dissipation
  - Extremely low → Suitable for applications using battery power

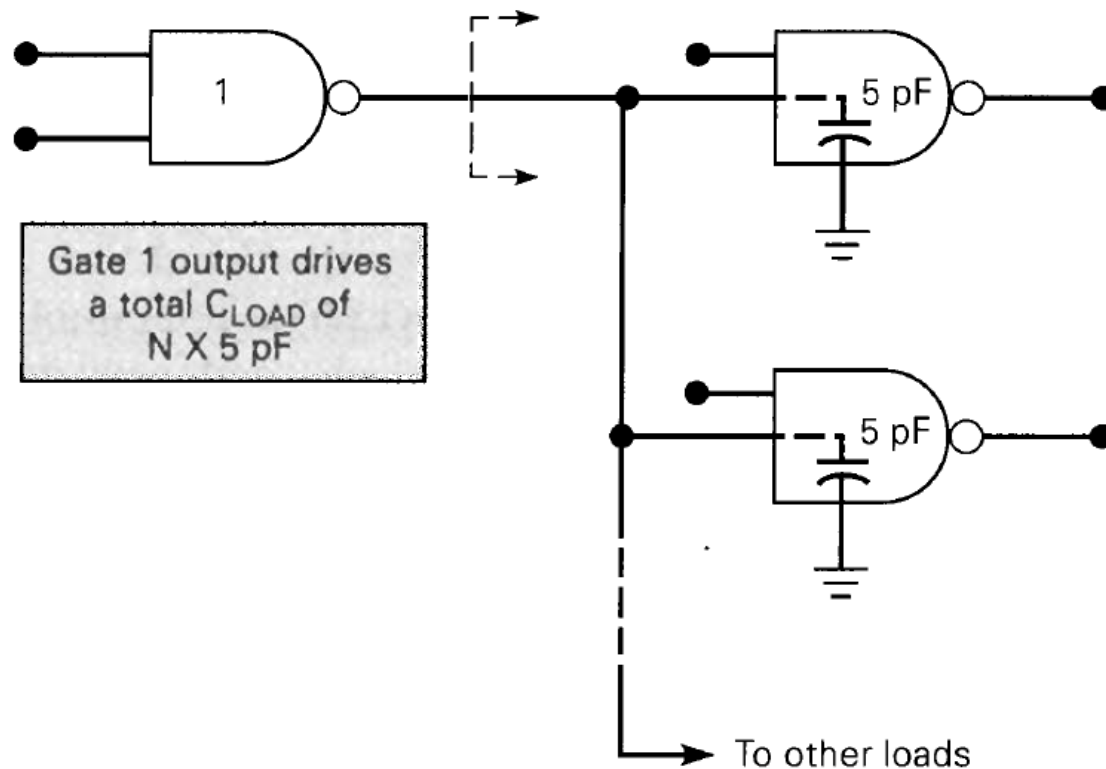
# CMOS Characteristics (2)

- $P_D$  increases with Frequency
  - Increase in proportion to the frequency at switching states of the circuit (LOW to HIGH)



# CMOS Characteristics (3)

- Fan-Out: depend on
  - Permissible maximum propagation delay
  - Frequency ???

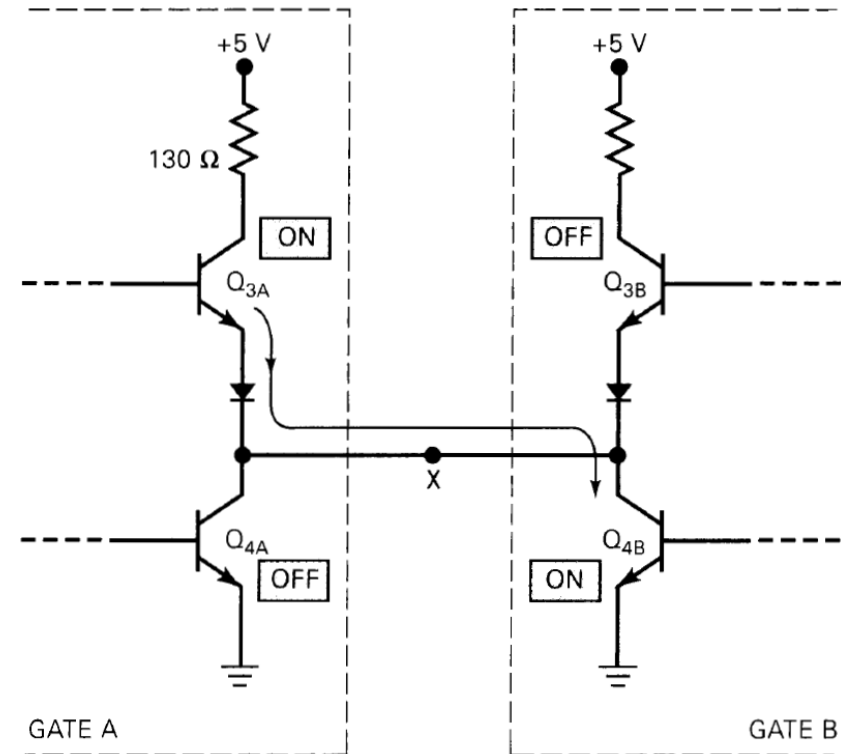
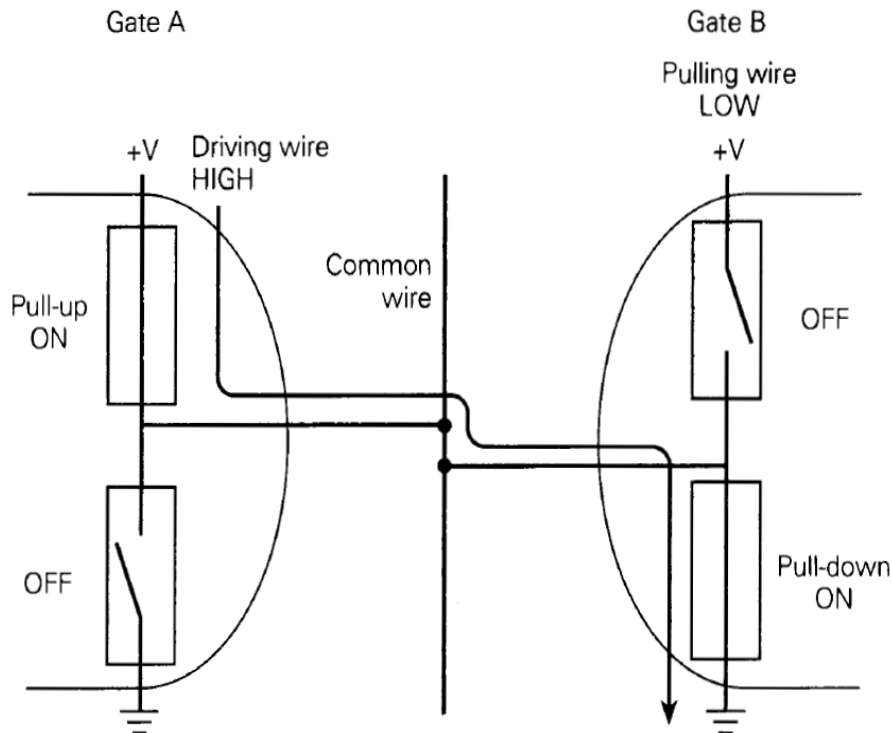


# CMOS Characteristics (4)

- Switching Speed
  - Faster than N-MOS and P-MOS (low output resistance
    - 1 k $\Omega$  vs. 100 k $\Omega$ )
- Unused Inputs
  - **Never left disconnected** : susceptible to noise and static charges
  - Tie to fixed voltage level (0V or  $V_{DD}$ ) or to another input
- Static Sentivity
- Latch-Up
  - Transistors stay ON permanently
  - Solution: Clamping diodes

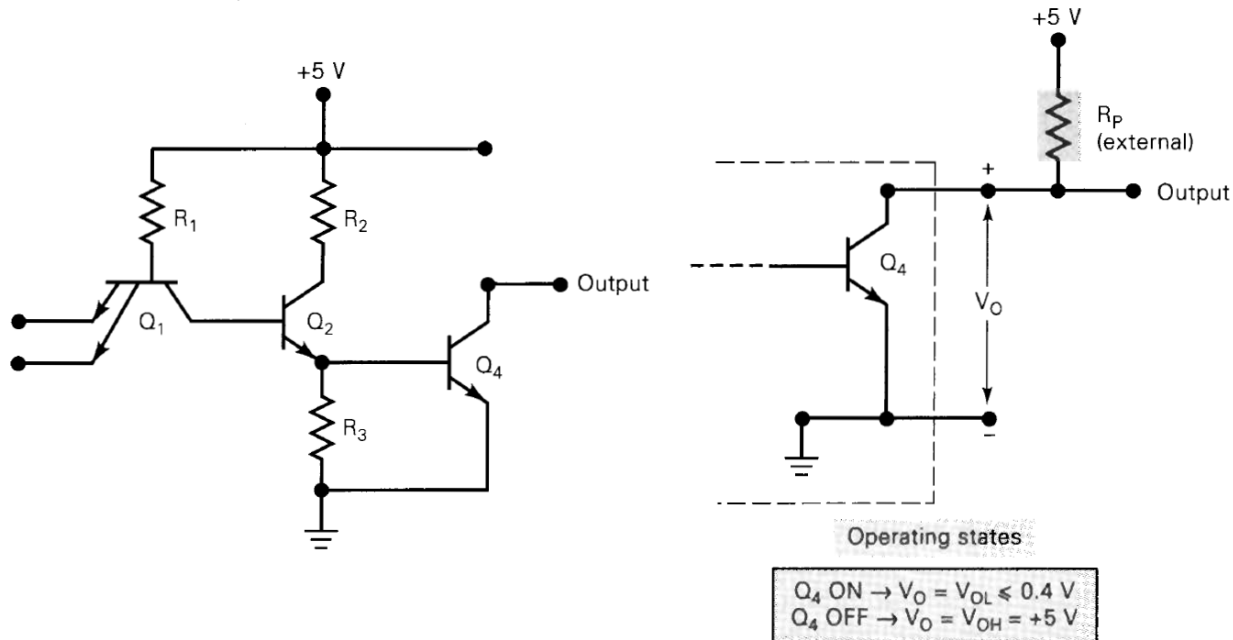
# Output Combination (sharing a common wire)

- Conventional CMOS outputs, TTL totem pole outputs should **never be tied together**
  - Output voltage: in the indeterminate range
  - Damage the ICs

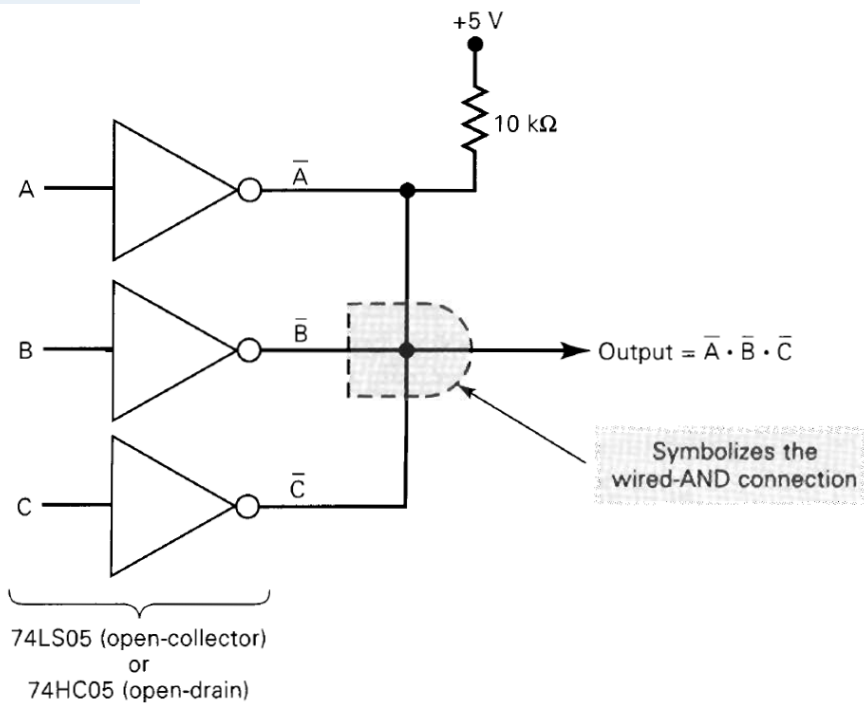


# Open-Collector/Open-Drain Outputs

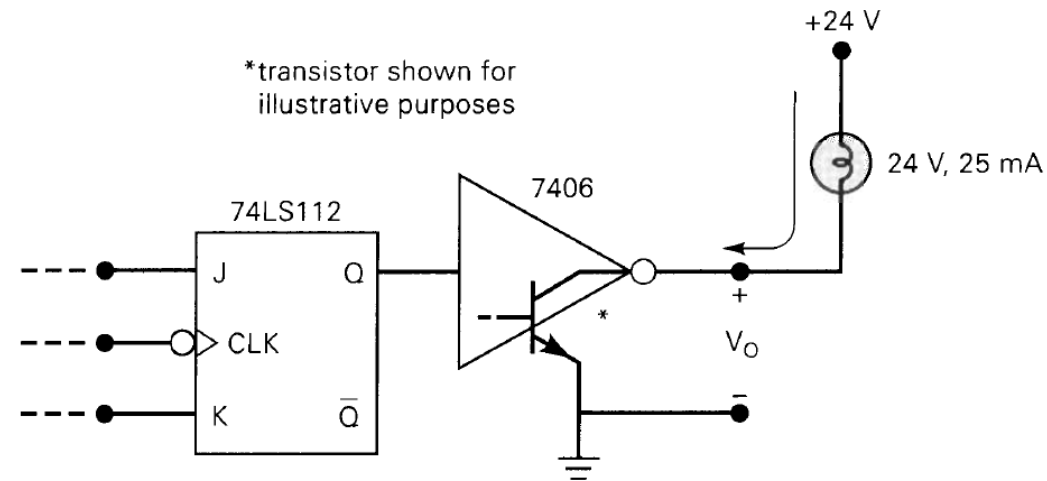
- Solution: **remove** the *active pull-up transistor*
  - CMOS output circuits: **open-drain outputs**
  - TTL totem pole output circuit: **open-collector outputs**
  - Output LOW state: no problem
  - Output HIGH state: *external pull-up resistor  $R_p$*



# Open-collector/drain outputs Application



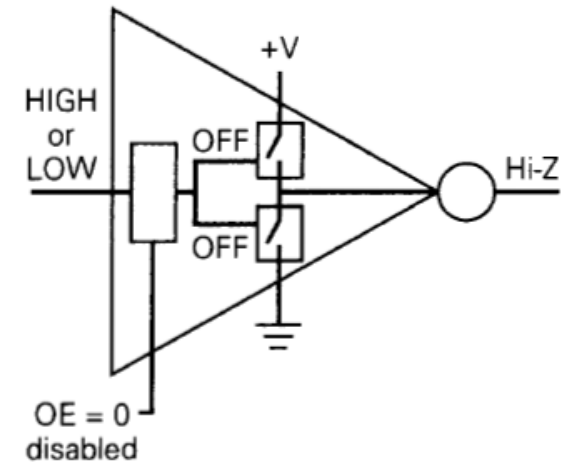
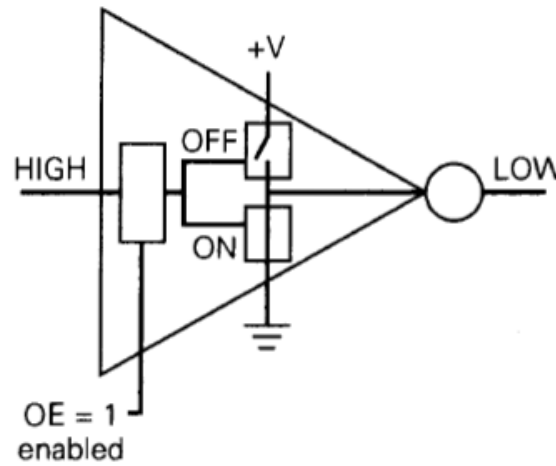
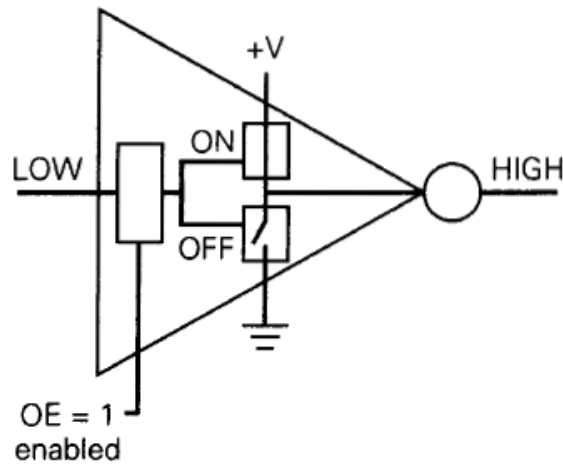
## Wired-AND connection



## Open-collector/drain Buffer/Driver

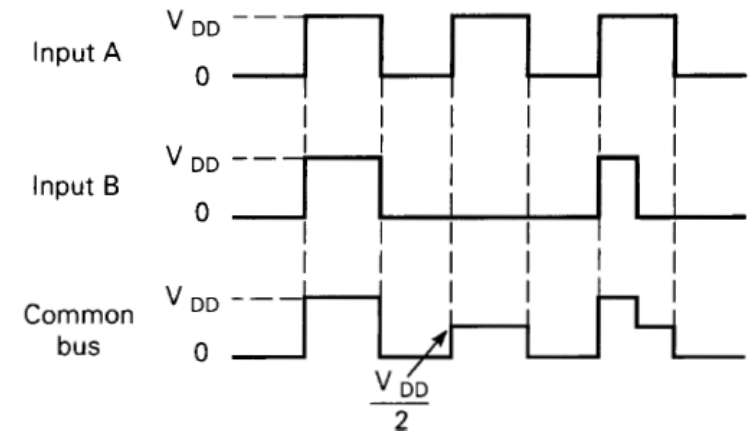
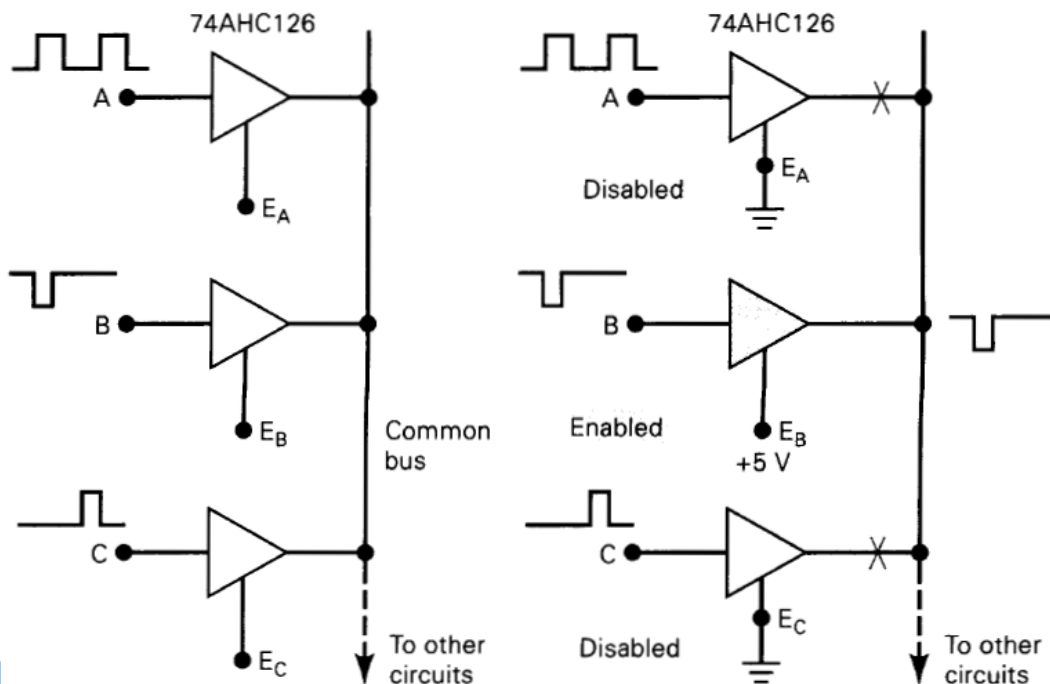
# Tristate (Three-State) Logic Outputs

- Allow three possible output states:
  - HIGH, LOW
  - High-impedance (Hi-Z): both pull-up and pull-down transistors are turned OFF
- Have an *enable* input (OE – output enable)



# Tristate (Three-State) Logic Outputs

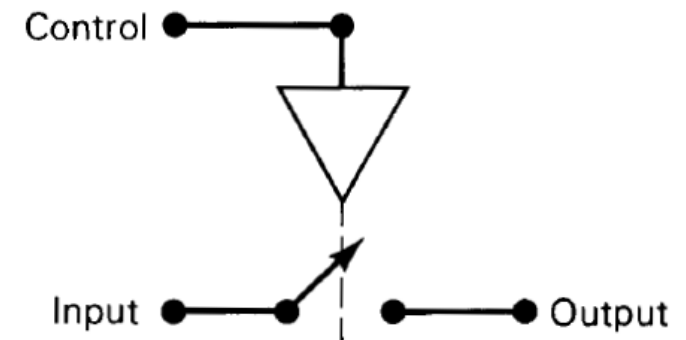
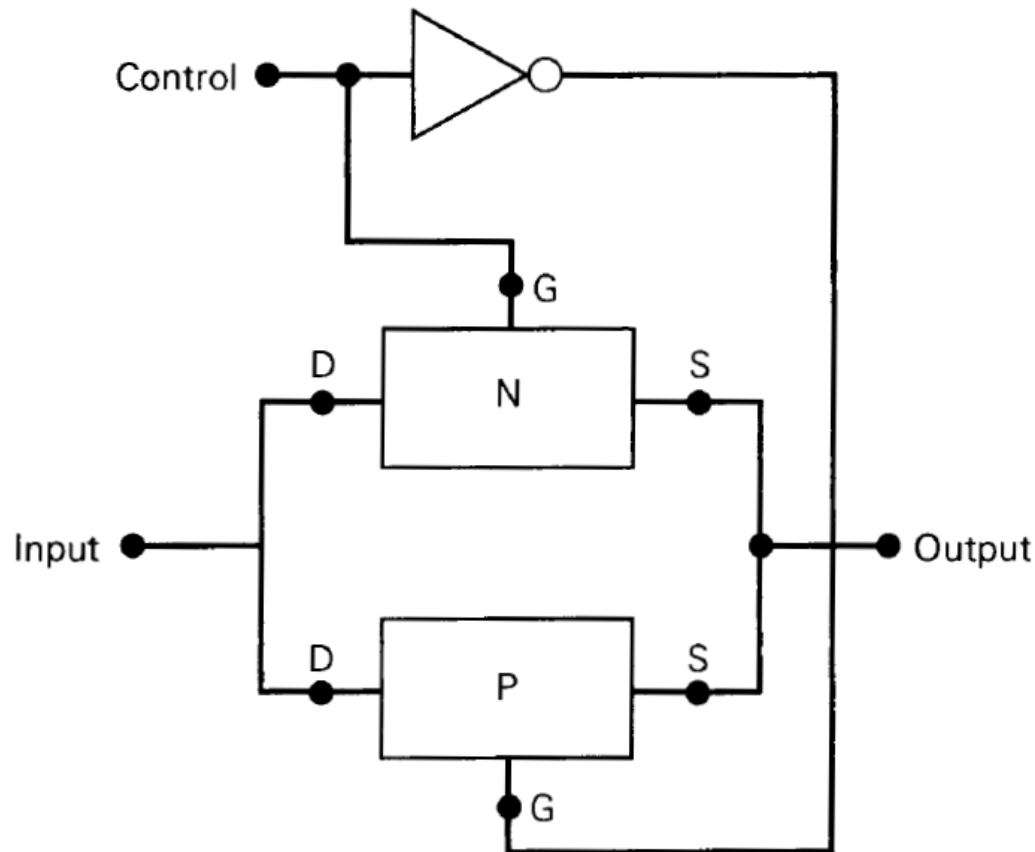
- Advantage
  - Tristate ICs output can be connected together
  - Low-impedance, high-speed characteristic
- Application: Tristate Buffers



Bus contention

# CMOS Transmission Gate (Bilateral Switch)

- Pass signals in both directions
- Digital and analog applications



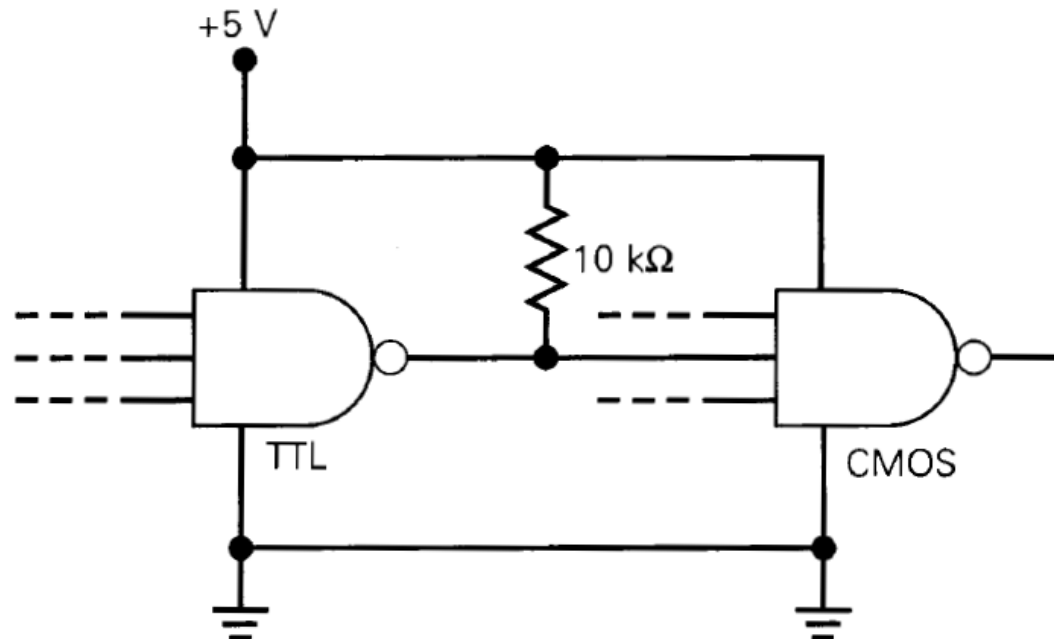
| Control input    | Switch      |
|------------------|-------------|
| 0 V              | Open (OFF)  |
| +V <sub>DD</sub> | Closed (ON) |

# IC Interfacing

- Connecting the output(s) to the input(s) of different electrical characteristic systems.
- Take advantage of the strong points of each IC family
  - High-speed, greater output current/voltage capability, high frequency
- Concerned problems
  - Fan-out
  - Voltage and current parameters

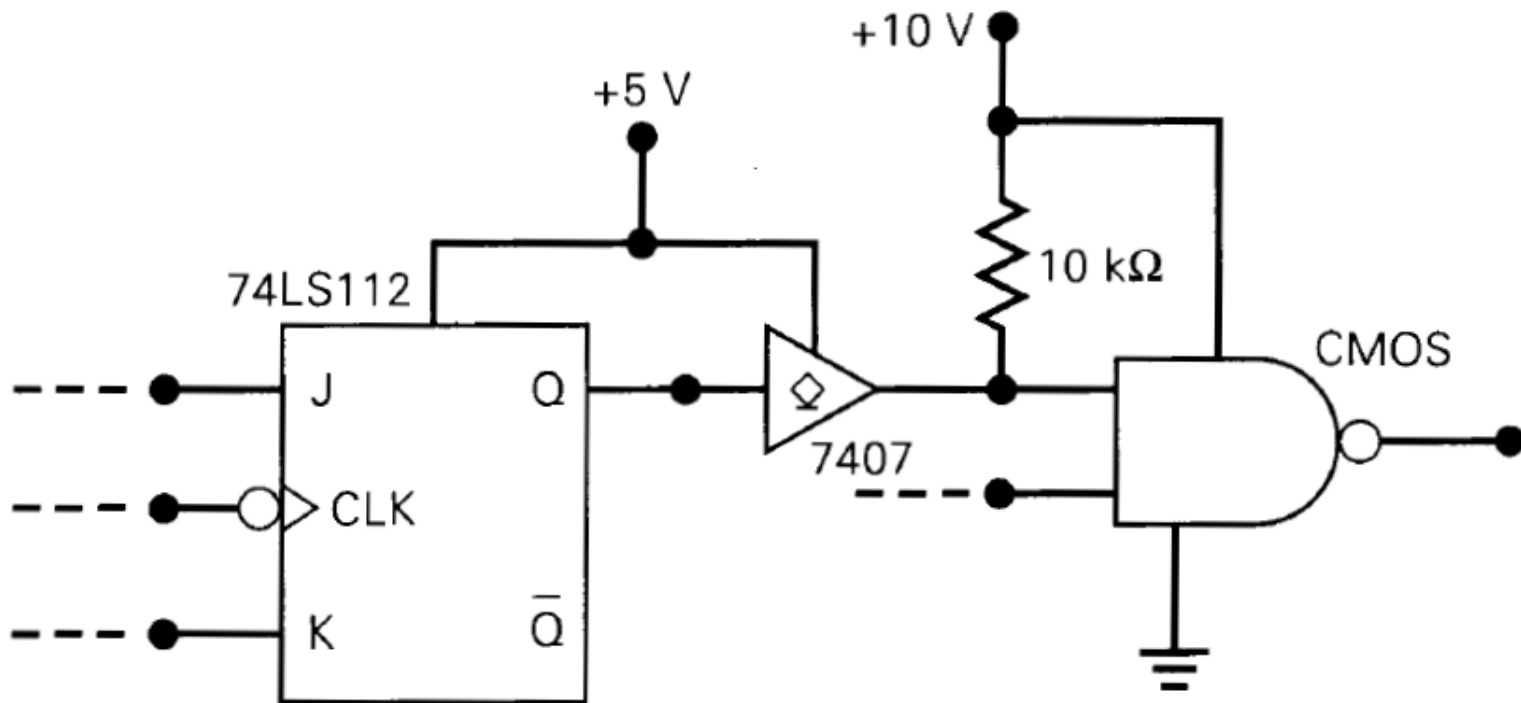
# TTL Driving CMOS

- Current: no problem
- Output voltages
  - $V_{OH}(\text{min})$  of TTL: too low when compared with  $V_{IH}(\text{min})$  of CMOS
  - Solution: *Pull-up resistor* at TTL output



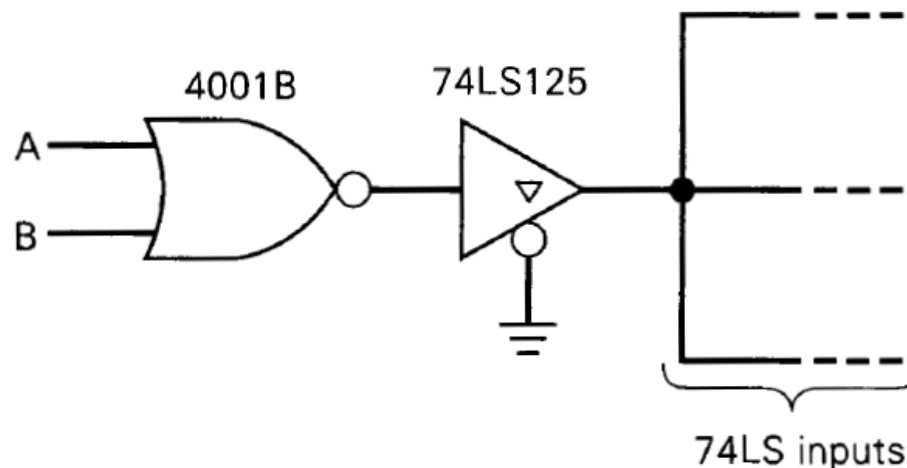
# TTL Driving CMOS

- Driving High-Voltage CMOS
  - Use TTL series from certain manufacturers that can operate with an high-voltage output pull-up
  - Utilize a **voltage level-translator**



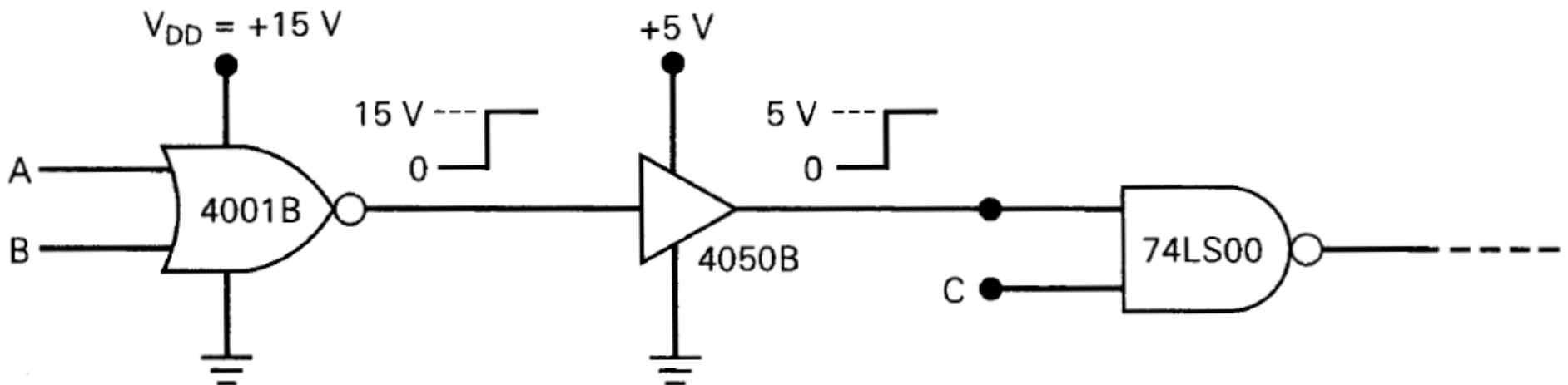
# CMOS Driving TTL

- Driving TTL in the HIGH State: no problem
- Driving TTL in the LOW State:
  - Excepting 4000B series, 74HC/74HCT series have no trouble driving a single TTL load of any series
  - Concern fan-out problem
  - Use buffer to interface low-current CMOS to high-current TTL



# CMOS Driving TTL

- High-Voltage CMOS Driving TTL
  - Use TTL series that can withstand high-voltage input
  - Use voltage level-translator



# Reference

- Chapter 8, Digital System – Principles and Applications, Ronald J.Tocci, Neal S. Widmer